

CALIFORNIA STATE UNIVERSITY, NORTHRIDGE

Design of X-Band Power Amplifier

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For the degree of Master of Science in

Electrical Engineering

By

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Abstract

Design of X-Band Power Amplifier

By

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Master of Science in Electrical Engineering

The objective of this project is to design a single-stage RF power amplifier operated at frequency of 9GHz in the X Band (8GHz to 12GHz) with a minimum output power of 37dBm at a minimum gain of 10dB. A Qorvo Discrete Power GaN on SiC HEMT transistor (TGF2023-2-02) in die form is chosen to meet design requirements. Using the transistor non-linear simulation model developed by Modelithics, the amplifier is designed and simulated in Agilent Advanced Design System (ADS). First, a DC bias circuit is designed to achieve required bias point at $V_{DS}=28V$ and $I_{DS}=125mA$. Next, power contours-plot (obtaining from simulated Load-Pull) method is used to design the amplifier. Input and output matching networks using microstrip lines are then designed to meet the design specification. Finally, the entire design is put together and simulated to show the overall amplifier performance.

Section 1: Introduction

In wireless communication system, RF power amplifier is normally located at the back end of the transmitter, right before the antenna to give the signal a boost needed to overcome the channel loss and interference. The required output power of an RF power amplifier ranges from a few milli-Watts to several hundred Watts and is determined by the application and the distance of the wireless link, the distance between the transmitter and the intended receiver. An RF power amplifier that is designed using a single active device (such as BJT or FET) is called a single-stage amplifier. A multi-stage amplifier is an amplifier designed with two or more active devices cascaded to increase power gain and output power and is widely used in applications requiring high output power. However, designing a multi-stage amplifier is more challenging and complex than designing a single-stage amplifier since active-loading between intermediate stages needs to be considered rather than just a passive-loading as in single-stage amplifier design. RF amplifiers are furthermore divided into classes (Class A, AB, B, C, etc...) based on the time duration (or conduction angle) that the transistor is in active region in a signal cycle. The transistor conduction angle is determined by DC bias point location. If the bias point allows the transistor to conduct for an entire signal cycle, the amplifier is called Class A amplifier. Amplifier with transistor conducted only for a half cycle is a Class B. In Class AB amplifier, the transistor conducts less than a full cycle but more than half of a cycle.

RF power amplifier design is typically optimized for maximum output power or efficiency rather than optimizing for power gain. RF power amplifier operating performance is specified in terms of the followings parameters:

- Frequency Range
- Output Power
- Power Gain
- Power-Added Efficiency

Section 2: Design Requirement

Just like other designs, the design of an RF power amplifier starts out with the design requirement that specifies the operating frequency, the desired gain and output power, a minimum power-added efficiency, and perhaps a tolerable distortion level. The power amplifier design requirement of this project is specified in the table below.

Frequency	9GHz
Output Power	> 37dBm
Power Gain	> 10dB

Table 1 – X-Band Power Amplifier Design Specification

To meet the design specification, TGF2023-2-02, a 12W Discrete Power GaN on SiC HEMT transistor is chosen. This transistor is rated for frequency range from DC to 18GHz, 41.2dBm nominal P_{SAT} at 6GHz, and 18dB linear gain at 6GHz. Full datasheet is in the Appendix A for additional information.

The transistor non-linear model from Modelithics Qorvo GaN Library is used in ADS to simulate the design. The Simulation Model is also in the appendix for review.

Section 3: Design Theory

A single-stage transistor amplifier consisted of a microwave transistor (BJT or FET) with its input and output matching networks is shown in Figure 1 below. The transistor, a two-port network, is characterized by a 2x2 S-parameters matrix [S] that tends to vary with frequency and input power.

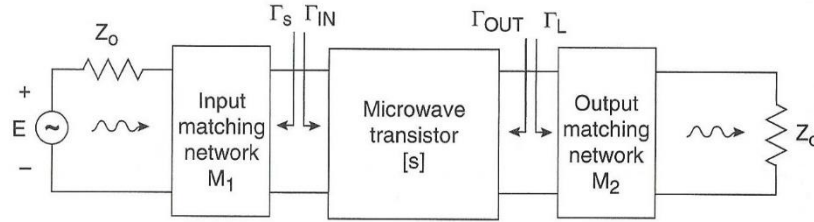


Figure 1 – An Amplifier Circuit with Matching Networks

RF power amplifier design process is to design input and output matching networks around the selected active device to meet design criteria. Source and load reflection coefficients are required to design the matching networks.

The input reflection coefficient Γ_{IN} and output reflection coefficient Γ_{OUT} are derived as:

$$\Gamma_{IN} = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1-S_{22}\Gamma_L} \quad (1)$$

$$\Gamma_{OUT} = S_{22} + \frac{S_{12}S_{21}\Gamma_S}{1-S_{11}\Gamma_S} \quad (2)$$

Where Γ_S and Γ_L are source and load reflection coefficients, respectively.

3.1 Stability

It can be seen from Equation (1) and (2) that if $|\Gamma_{IN}| > 1$ or $|\Gamma_{OUT}| > 1$ (i.e. reflected signal power is greater than incident signal power) the transistor amplifier becomes unstable. A transistor is considered Unconditionally Stable if $|\Gamma_{IN}| < 1$ and $|\Gamma_{OUT}| < 1$ for all passive source and load impedance (with $|\Gamma_S| < 1$ and $|\Gamma_L| < 1$). Otherwise, it is said to be Conditionally Stable if only a certain passive source and load impedance can make $|\Gamma_S| < 1$ and $|\Gamma_L| < 1$.

A network is Unconditionally Stable only if one of the following criteria is met:

1. Three-Parameter Test

$$K > 1, \text{ and } \frac{1-|S_{11}|^2}{|S_{12}S_{21}|} > 1, \text{ and } \frac{1-|S_{22}|^2}{|S_{12}S_{21}|} > 1$$

2. K- Δ Test

$$K > 1 \text{ and } |\Delta| < 1$$

3. K- B_1 Test

$$K > 1 \text{ and } B_1 > 0$$

$$\text{Where } \Delta = S_{11}S_{22} - S_{12}S_{21}, K = \frac{1-|S_{11}|^2-|S_{22}|^2+|\Delta|^2}{2|S_{12}S_{21}|},$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2$$

The input and output matching networks affect source reflection coefficient Γ_S and load reflection coefficient Γ_L that determine the stability of the transistor amplifier. If a network is Conditionally Stable, it is only stable for a certain source and load impedances

and thus for such devices, input and output matching networks design choices are constrained and limited.

3.2 Power Gain

Amplifier power gain is another important design parameter that is considered in the design process. Power gain is defined as the ratio of two power levels.

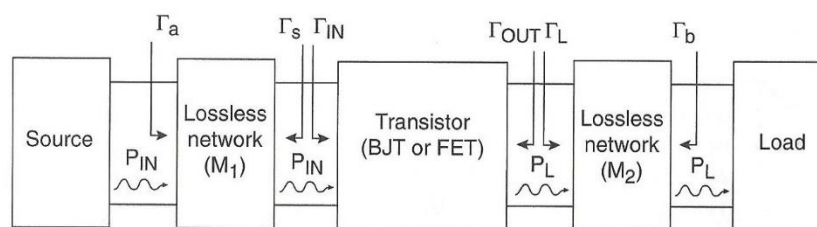


Figure 2 – A transistor amplifier with lossless networks

Consider a transistor amplifier with lossless matching networks as shown in Figure 2, three different power gains are defined as follow:

3.2.1 Transducer Power Gain

Transducer Power Gain is defined as the ratio of the power delivered to the load P_L to the power available from the source P_{AVS}

$$G_T = \frac{P_L}{P_{AVS}}$$

G_T is derived to be:

$$G_T = \frac{1-|\Gamma_S|^2}{|1-\Gamma_{IN}\Gamma_S|^2} |S_{21}|^2 \frac{1-|\Gamma_L|^2}{|1-S_{22}\Gamma_L|^2} \quad (\text{in term of } \Gamma_{IN}) \quad (3)$$

$$G_T = \frac{1-|\Gamma_S|^2}{|1-S_{11}\Gamma_S|^2} |S_{21}|^2 \frac{1-|\Gamma_L|^2}{|1-\Gamma_{OUT}\Gamma_L|^2} \quad (\text{in term of } \Gamma_{OUT}) \quad (4)$$

3.2.2 Operating Power Gain

Operating Power Gain is defined as the ratio of the power delivered to the load P_L to the power input to the transistor P_{IN}

$$G_P = \frac{P_L}{P_{IN}}$$

G_P is derived to be:

$$G_P = \frac{1}{1 - |\Gamma_{IN}|^2} |S_{21}|^2 \frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2}$$

3.2.3 Available Power Gain

Available Power Gain is defined as the ratio of the power available from the network (transistor) P_{AVN} to the power available from the source P_{AVS}

$$G_A = \frac{P_{AVN}}{P_{AVS}}$$

G_A is derived to be:

$$G_A = \frac{1 - |\Gamma_S|^2}{|1 - S_{11}\Gamma_S|^2} |S_{21}|^2 \frac{1}{1 - |\Gamma_{OUT}|^2}$$

Note that $P_{AVS} = P_{IN}$ when $\Gamma_{IN} = \Gamma_S^*$ (input matched condition) and $P_{AVN} = P_L$ when $\Gamma_{OUT} = \Gamma_L^*$ (output matched condition). Thus,

$$G_P = G_T \text{ (input matched condition)}$$

$$G_A = G_T \text{ (output matched condition)}$$

It can be observed from Equation (3) that the Transducer Power Gain G_T is the gain product of the effective gain of the input matching network G_S , the transistor gain G_o , and the effective gain of the output matching network G_L .

$$G_T = G_S G_o G_L$$

Where

$$G_o = |S_{21}|^2$$

$$G_S = \frac{1 - |\Gamma_S|^2}{|1 - \Gamma_{IN} \Gamma_S|^2}$$

$$G_L = \frac{1 - |\Gamma_L|^2}{|1 - S_{22} \Gamma_L|^2}$$

3.3 Power Added Efficiency

Power amplifier is a main source of power consumption in a transmitter. RF power amplifier consumes DC power to generate RF power (and heat). It is highly desired to have most (if not all) of the DC power convert to RF power by the amplifier. How efficient an RF power amplifier converts DC power to RF power is quantified by Power Added Efficiency.

Power Added Efficiency (PAE) is defined in [4] as:

$$PAE = \frac{P_{AVN} - P_{IN}}{P_{DC}}$$

Where P_{DC} is DC power consumption

3.4 Unilateral

When the transistor reverse transmission coefficient S_{12} is zero, the input reflection coefficient Γ_{IN} and output reflection coefficient Γ_{OUT} are no longer affected by the load reflection coefficient Γ_L and source reflection coefficient Γ_S , respectively. Equation (1) and Equation (2) are reduced to

$$\Gamma_{IN} = S_{11}$$

$$\Gamma_{OUT} = S_{22}$$

Under conjugate matched condition, the source and load reflection coefficient are now independent on each other.

$$\Gamma_S^* = \Gamma_{IN} = S_{11}$$

$$\Gamma_L^* = \Gamma_{OUT} = S_{22}$$

The transistor has effectively de-coupled its input and output reflection coefficients and thus simplified power amplifier design process when the transistor is unilateral ($S_{12} = 0$).

3.4.1 Unilateral Figure of Merit

Even if transistor $S_{12} \neq 0$, unilateral design can still be considered if the Unilateral Figure of Merit U yields an acceptable gain error due to unilateral assumption. The maximum error is bounded by:

$$\frac{1}{|1 + U|^2} < \frac{G_T}{G_{TU,max}} < \frac{1}{|1 - U|^2}$$

$$\text{Where } U = \frac{|S_{12}||S_{21}||S_{11}||S_{22}|}{(1-|S_{11}|^2)(1-|S_{22}|^2)}$$

And $G_{TU,max}$ is maximum unilateral Transducer Gain

3.5 Matching Networks

In general, matching networks are used to transform one impedance to another impedance. In RF amplifier design, matching networks are used to tune input and output reflection coefficients to the values needed to achieve desired design goals such as stability, gain, and output power. Matching networks can be design using lumped or distributed elements. Distributed matching networks have much lower loss than their lumped counterpart and are often used at high frequencies.

3.5.1 Distributed Matching Networks

Distributed matching networks are designed using microstrip or stripline.

Microstrips have copper loss, dielectric loss, and radiation or conduction loss.

However, microstrip insertion loss is small and often neglectable.

3.6 RF Power Amplifier Design

Based on input signal level, RF amplifiers are classified into two types: Small-Signal amplifiers and Large-Signal amplifiers. Small-Signal amplifiers such as Low-Noise amplifier LNA, High-Gain amplifier HGA, and Maximum-Gain amplifier MGA are resided in RF receiver's front end to amplify coming weak (small) and noisy signals whereas Large-Signal amplifiers are power amplifiers located at back end of the transmitter to increase signal power for transmitting. Large-signal amplifier design which is also known as power amplifier design is significantly different from the design of small-signal amplifier. Small-signal amplifier design uses small-signal S-parameters which can be easily measured or readily supplied by the manufacturer. Small-signal S-parameters cannot be used in designing large-signal amplifier where signals go beyond

linear operating region of the transistor. Large-signal S-parameters are normally not available due to test equipment limited power handling and due to transistor's stability during characterization (input/output opened or shorted). However, there are other methods to design a large-signal amplifier that are listed in [1] and as shown below.

3.6.1 The Modified S-parameters Method

A set of modified S-parameters can be used to design a class A power amplifier. The modified S-parameters are almost the same as the small-signal S-parameters except with a reduced forward transmission gain S_{21} . The lower S_{21} is due to output saturation of the transistor at high power signal level.

3.6.2 The Source and Load Reflection Coefficients Method

The source and load reflection coefficients are data measured at when the output of the transistor is at 1-dB gain compression point and specified by the manufacturer. 1-dB compression point is the point where power gain is reduced by 1dB from the small-signal gain due to non-linearities.

3.6.3 Power Contours-Plot Method

Power contours-plot is the mapping of transistor output power on the Smith Chart at different load impedance (or load reflection coefficient) while providing a matched input impedance (conjugately matched) for maximum power transfer.

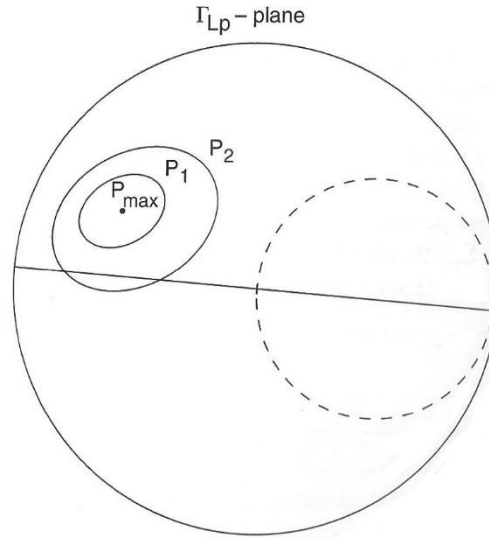


Figure 2 – Power Contours-Plot

Power contours have oval shapes and not circles due to transistor non-linearity at large signal condition. Points along a contour have same output power and each contour represents a power level.

Power Contours-Plot method design procedure is listed below:

- Step 1: Locate a power contour on the plot that is corresponding to the desire output power level. If needed, interpolate and draw to determine power contour between two available known power contours
- Step 2: From the selected power contour, select load reflection coefficient $\Gamma_L = \Gamma_{LP}$ point on the contour and in the stable region
- Step 3: Calculate input reflection coefficient Γ_{IN} based on selected Γ_L
- Step 4: Set Γ_S to Γ_{IN}^* for maximum power transfer
- Step 5: Design matching networks to obtain selected Γ_S and Γ_L

Ultimately, RF power amplifier design involves choosing an appropriate transistor and designing input and output matching networks for that transistor to make it capable of delivering desired output power and gain along with meeting other design criteria. In this design project Power Contours-Plot method will be used to design the power amplifier.

Section 4: The Design

To meet the design goal, a Qorvo power transistor TGF2023-2-02 (data sheet in Appendix A) is selected. The transistor has a frequency range from DC to 18GHz and a saturated output power of 41.2dBm with a power gain of 14.9dB at 6GHz. The transistor performance is sufficient to design an amplifier with the following requirements:

- Operating frequency: 9GHz
- Minimum output power: 37dBm
- Minimum power gain: 10dB

The TGF2023-2-02 transistor when biased at $V_{DS} = 28V_{DC}$ and $I_{DS} = 125mA$ has the following measured S-parameters at 9GHz:

$$S_{11} = -0.855 + j0.34383 = 0.9212 \angle 158.1^\circ$$

$$S_{12} = 0.01345 + j0.01768 = 0.0222 \angle 52.7^\circ$$

$$S_{21} = 1.50948 + j0.4026 = 1.5622 \angle 14.9^\circ$$

$$S_{22} = -0.63128 - j0.15806 = 0.6508 \angle -165.9^\circ$$

The design is done by simulation in ADS with transistor non-linear model developed by Modelithics. The design process is started with the determination of the gate-to-source voltage V_{GS} needed to set the transistor to the desired bias-point. Next, the transistor stability is checked. Load-Pull simulation is then performed to plot the power contours. Since the amplifier is designed to deliver maximum output power, the load reflection coefficient Γ_L is selected at a point on the maximum power contour. Using the select Γ_L ,

input reflection coefficient Γ_{IN} is then calculated. Source reflection coefficient Γ_S is set equal to Γ_{IN}^* for maximum power transfer. After Γ_S and Γ_L is determined, input and output matching networks are designed. Finally, the complete transistor amplifier design including the transistor, input/output matching networks and DC bias circuit, is simulated and its overall performance is presented.

4.1 V_{GS} Determination

The gate-source voltage V_{GS} required to bias the transistor at $V_{DS}=28V$ and $I_{DS}=125mA$ is determined by simulation with the setup below.

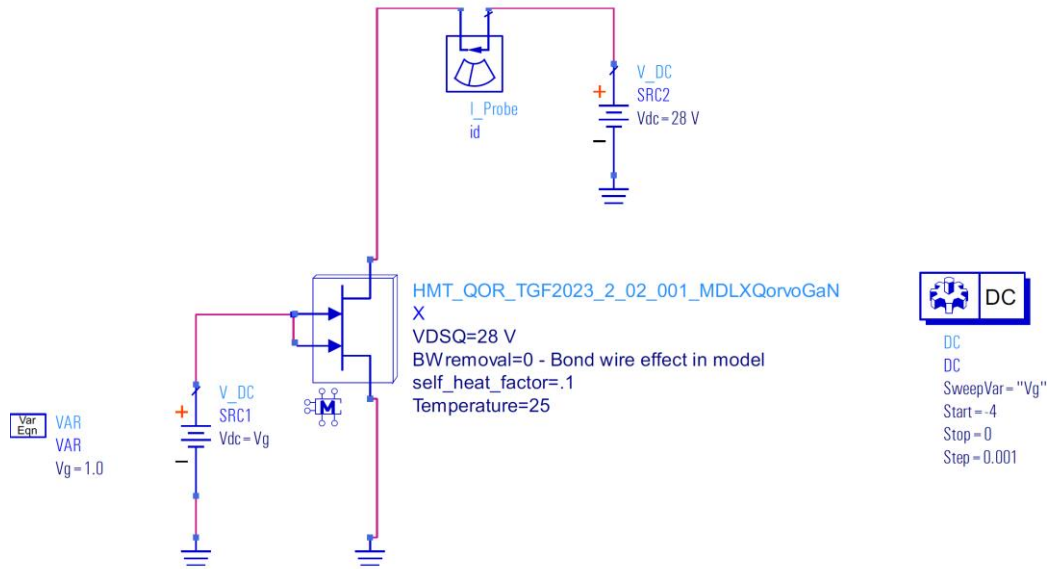


Figure 3 – V_{GS} Determination Setup

With V_{DS} set at $28V_{DC}$, V_{GS} is swept from $-4V$ to $0V$ in steps of $1mV$ and I_{DS} is recorded at each V_{GS} value. The result is plotted and the corresponding V_{GS} value that yields $I_{DS}=125mA$ is determined.

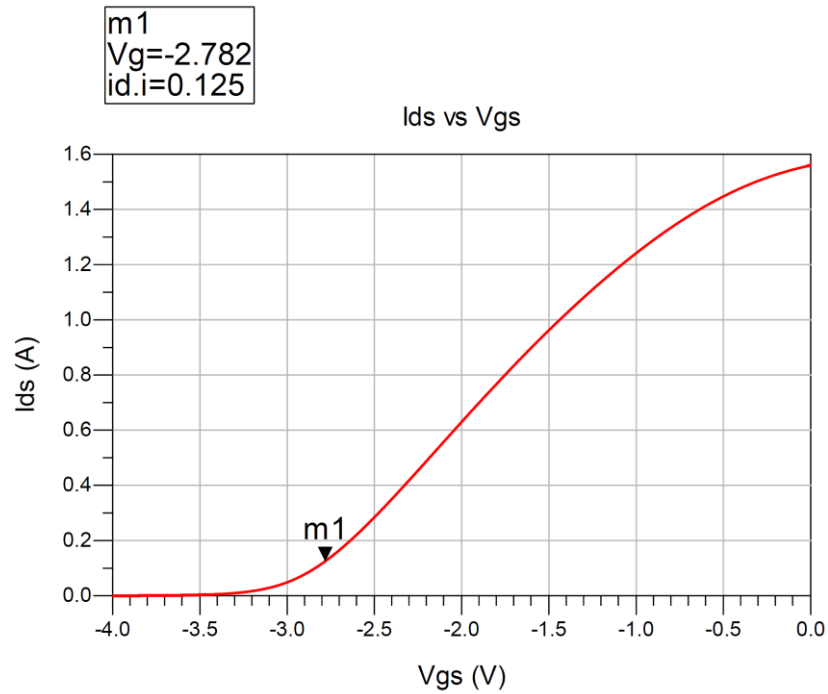


Figure 4 – Bias Point Result

As indicated by marker m1 in Figure 3, I_{DS} of $125mA$ is achieved with V_{GS} set at $-0.782V$.

4.2 S-Parameter Simulation

Transistor S-parameters can be extracted from the transistor model in ADS. The simulation schematic is shown below. RF chokes (DC Feeds) and DC blocks are ideal

component required for simulation purpose. Input and output ports are terminated with 50Ω . The simulation is run with frequency swept from 8GHz to 12GHz.

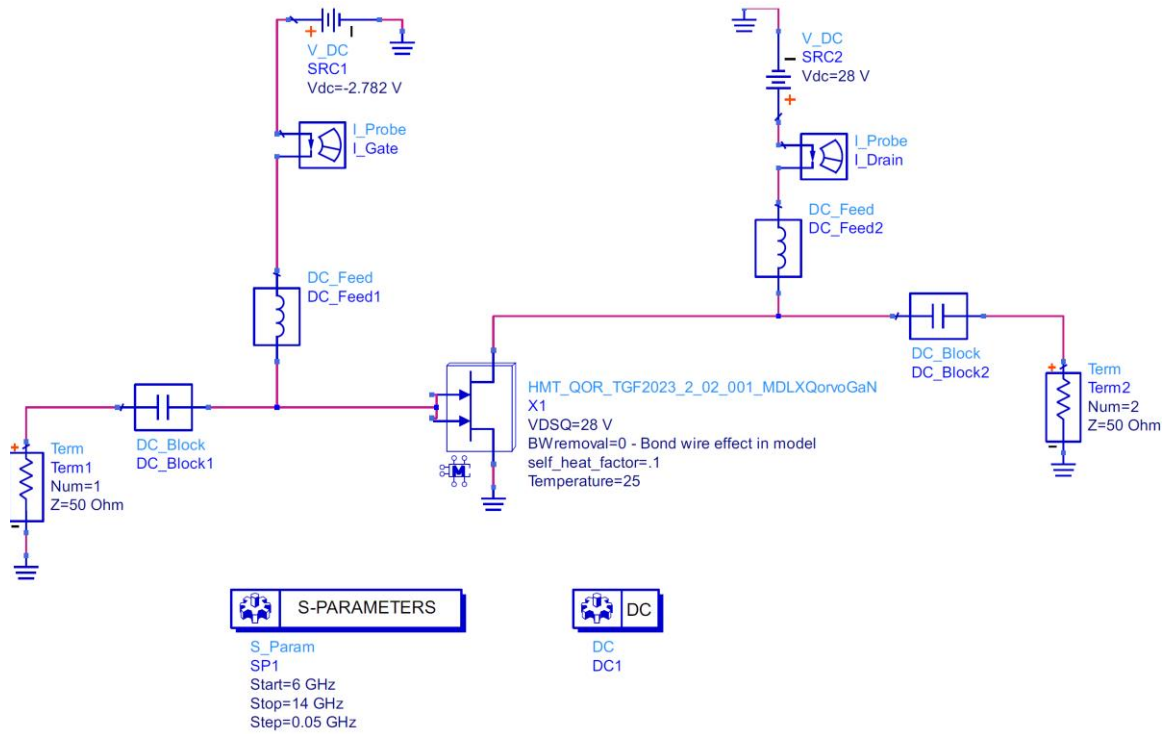


Figure 5 - S-Parameter Simulation Schematic

Simulated S-parameters from 8GHz to 12GHz is plotted. As expected, S_{21} decreases as frequency increases.

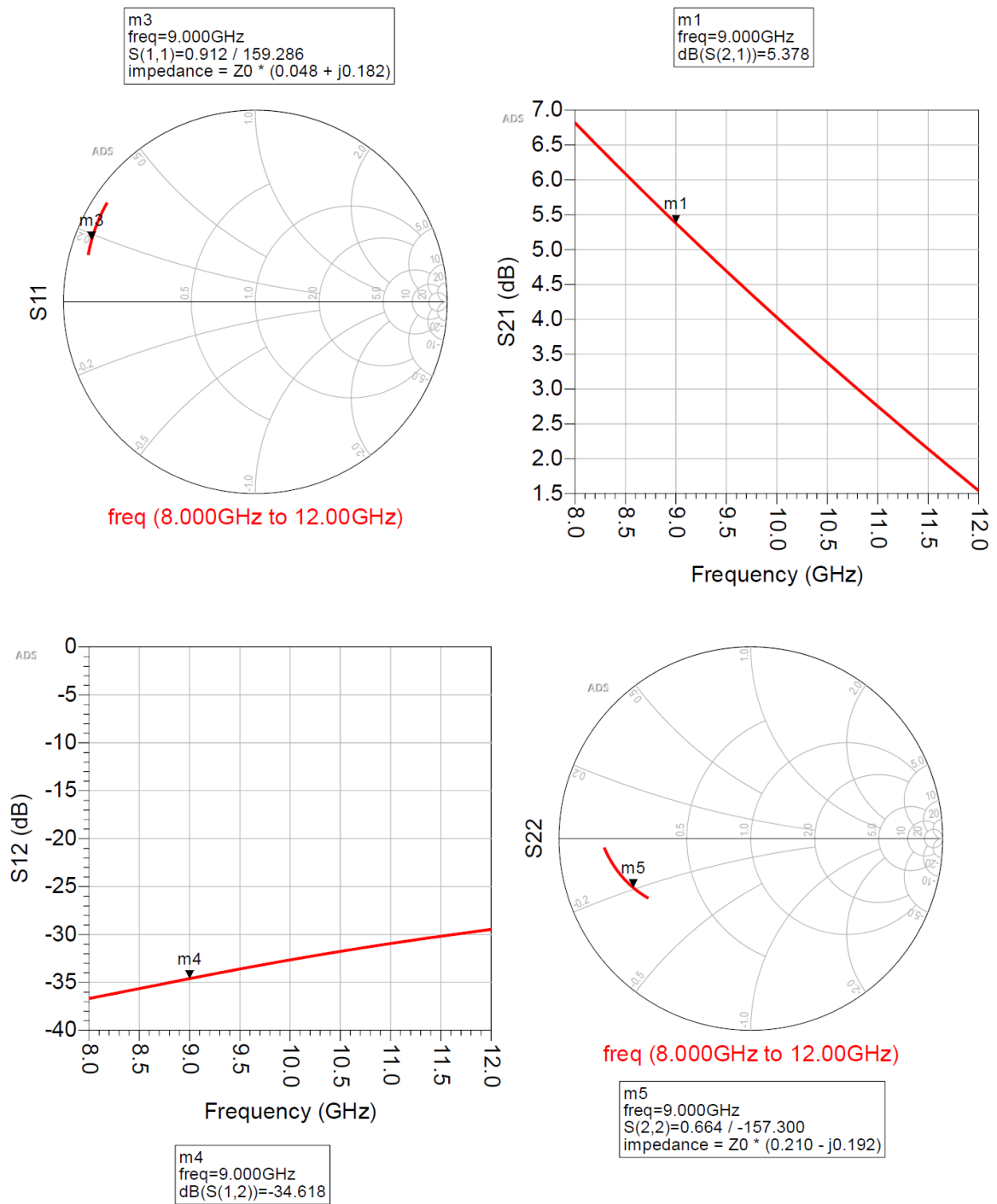


Figure 6 - Simulated S-Parameters vs. Frequency Plot

4.3 Stability Check

Before continuing further in designing the RF power amplifier, the stability of the transistor needs to be checked. K-B₁ criteria is used for stability check.

Using the measured S-parameters, K and B₁ are calculated to be:

$$|\Delta| = |S_{11}S_{22} - S_{12}S_{21}| = 0.592$$

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|} = 1.12$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2 = 1.07$$

Since $K > 1$ and $B_1 > 0$ the transistor is unconditionally stable.

4.4 Unilateral Figure of Merit

Since the transistor S_{12} is not zero, the Figure of Merit and gain error range need to be calculated to determine if unilateral design assumption can be used.

Electronic Waves version 2.1 software is used to calculate unilateral figure of merit and gain error range. U is found to be 0.23824 and the lower and upper error limits are -1.856dB and 2.364dB, respectively. Since the errors are large, the design is considered a bilateral design.

$$U = \frac{|S_{12}||S_{21}||S_{11}||S_{22}|}{(1 - |S_{11}|^2)(1 - |S_{22}|^2)} = 0.23824$$

$$-1.856dB = \frac{1}{|1 + U|^2} < \frac{G_T}{G_{TU,max}} < \frac{1}{|1 - U|^2} = 2.364dB$$

EXAMPLE 13.3		
What is the maximum error range if we use unilateral gain equations for a transistor having the following S- parameters?		
S_{11}	= 0.6	$\angle -160^\circ$
S_{21}	= 2.5	$\angle 30^\circ$
S_{12}	= 0.045	$\angle 16^\circ$
S_{22}	= 0.5	$\angle -90^\circ$



SOLUTION:		
INPUT DATA		
Parameter	Mag.	Angle
S_{11}	0.9212	158.1°
S_{21}	1.5622	14.9°
S_{12}	0.0222	52.7°
S_{22}	0.6508	-165.9°

OUTPUT DATA	
$U = \frac{ S_{12} S_{21} S_{11} S_{22} }{(1 - S_{11} ^2)(1 - S_{22} ^2)}$	
$U = 0.23824$	
$\frac{1}{(1+U)^2} < \frac{G_T}{G_{TU,max}} < \frac{1}{(1-U)^2}$	
The lower limit is given by:	
$1/(1+U)^2 = 0.652 = -1.856 \text{ dB}$	
And the upper limit is given by:	
$1/(1-U)^2 = 1.723 = 2.364 \text{ dB}$	
Therefore the maximum error range for G_T is:	
$-1.856 < G_T/G_{TU,max} < 2.364 \text{ dB}$	



Figure 7 – Unilateral Figure of Merit and Error Calculations

4.5 Load-Pull Simulation and Load Reflection Coefficient Selection

Load-Pull simulation is used to plot the power contours of the transistor. A point on the maximum output power contour is selected as the load reflection coefficient (or load impedance seen by the transistor). The load-pull setup is shown in Figure 8.

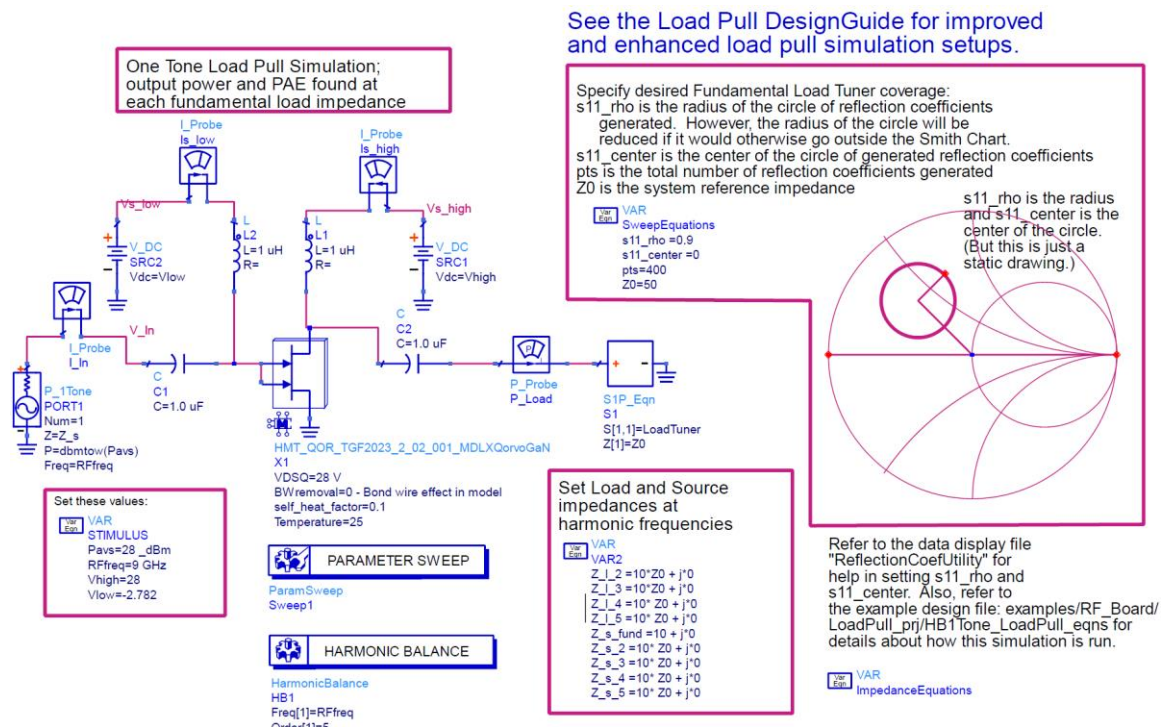


Figure 8 – Load-Pull Simulation Setup

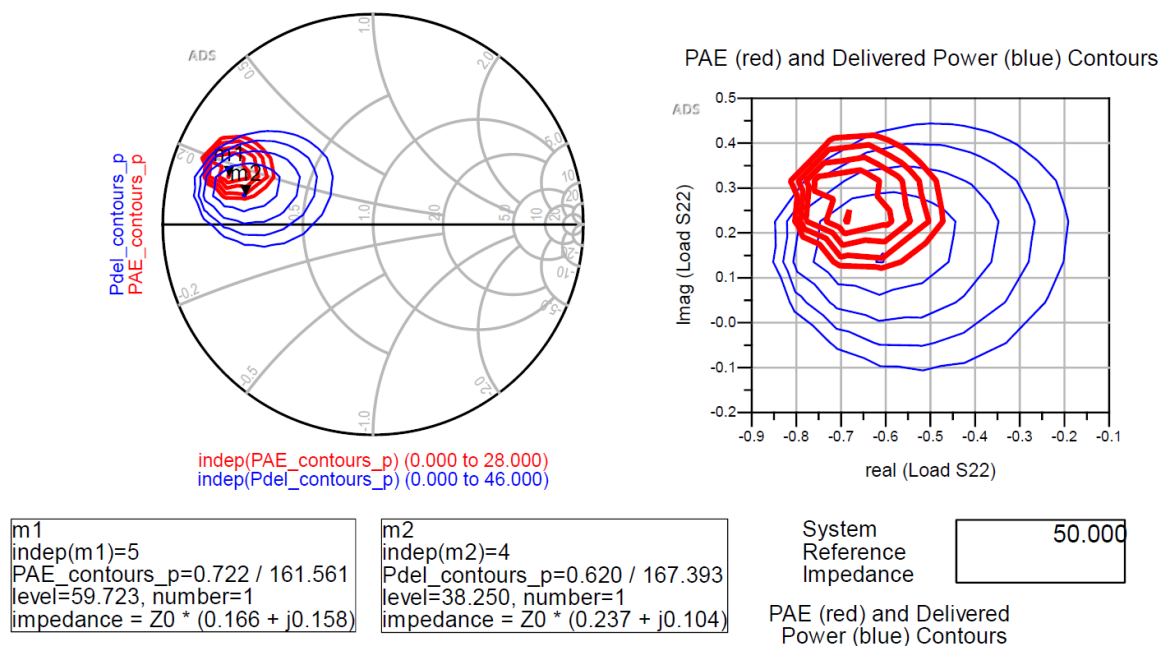


Figure 9 – Power Contours (Blue) and PAE (Red)

From the power contours plot and as indicated by marker m2, the transistor is capable of outputting 38.25dBm (gain of 10.25dB since input power $P_{IN} = 28\text{dBm}$) when the load reflection coefficient Γ_L is approximately equal to $-0.6 + j0.15$.

Load reflection coefficient Γ_L is selected to be at $-0.688 + j0.226$ ($Z_L=8.207+j7.779$) as pointed to by marker m3 in the figure below. The corresponding input reflection coefficient Γ_{IN} is also calculated and displayed in the figure. The resulting output power delivered is calculated to be 38.14dBm with power-added efficiency PAE of 59.82%.

$$PAE = \frac{P_L - P_{AVS}}{P_{DC}}$$

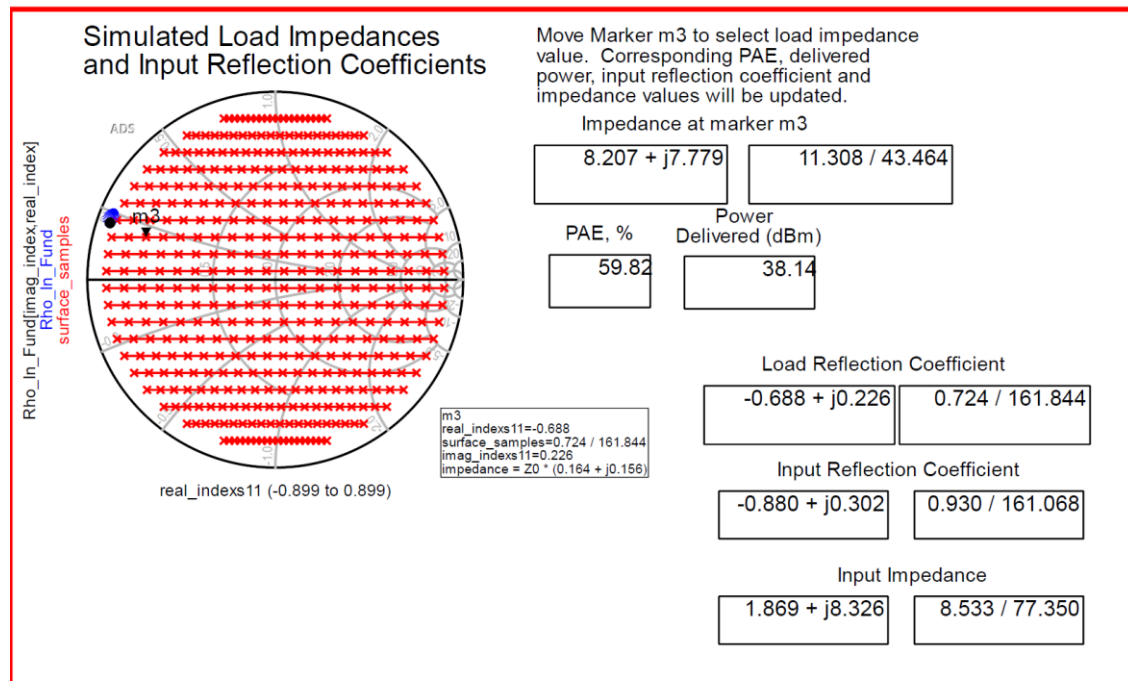


Figure 10 – Simulated Load and Input Reflection Coefficients

4.6 Input Reflection Coefficient Calculation

After Γ_L is selected, input reflection coefficient Γ_{IN} is determined using Equation (1).

$$\Gamma_L = -0.688 + j0.226$$

$$\Gamma_{IN} = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L} = -0.870 + j0.302$$

$$Z_{IN} = Z_0 \frac{1 + \Gamma_{IN}}{1 - \Gamma_{IN}} = 2.114 + j8.416$$

Hand calculation result is compared with results obtaining from ADS and Electronic Waves 2.1 software. They are closely matched.

	ADS (Simulation)				Electronic Waves 2.1 (S-Parameters)				My Excel SW (S-Parameters)				Hand Calculations (S-Parameters)			
	Re	Im	Mag	Degrees	Re	Im	Mag	Degrees	Re	Im	Mag	Degrees	Re	Im	Mag	Degrees
Γ_{IN}	-0.877	0.297	0.926	161.32			0.921	160.9	-0.870	0.302	0.921	160.9	-0.870	0.302	0.921	160.9
Z_{IN}	1.973	8.212	8.446	76.493					2.119	8.403	8.666	75.9	2.114	8.416	8.677	75.9

Figure 11 – Γ_{IN} Calculation Verification

The next steps are to design output and input matching networks to obtain:

4.7 Output Matching Network

The output matching network is designed to transform 50Ω nominal load to desired load impedance $Z_L = 8.207 + j7.779\Omega$ ($\Gamma_L = -0.688 + j0.226$) at which the transistor outputs maximum power. The design is carried out in ADS Smith Chart as shown in figures below. The matching network is an L-network having an open shunt stub with electrical length of 64.57° and a series stub of 30.81° .

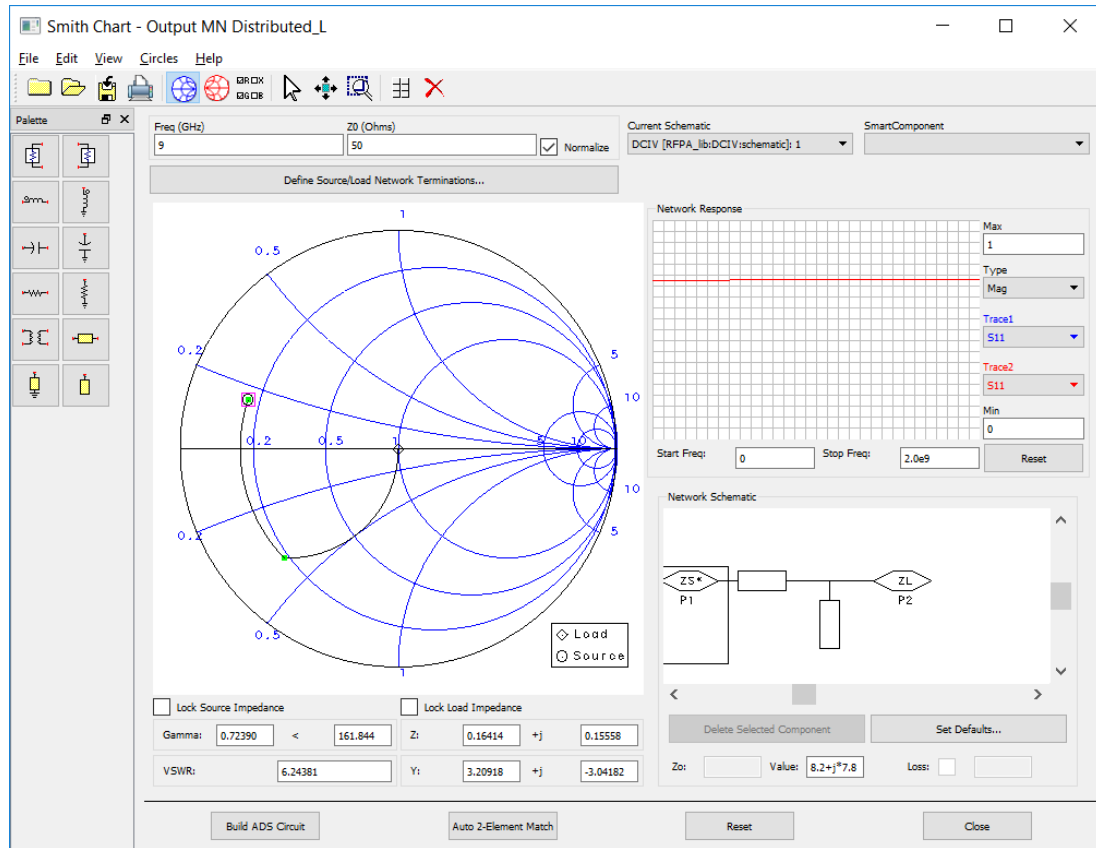


Figure 12 – Output Matching Network Design

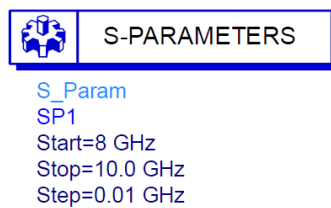
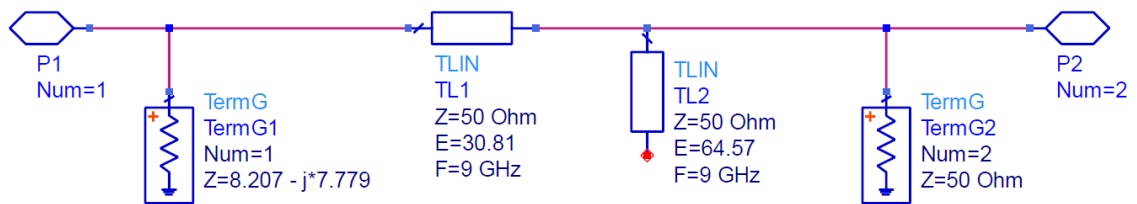


Figure 13 – Output Matching Network Design Schematic

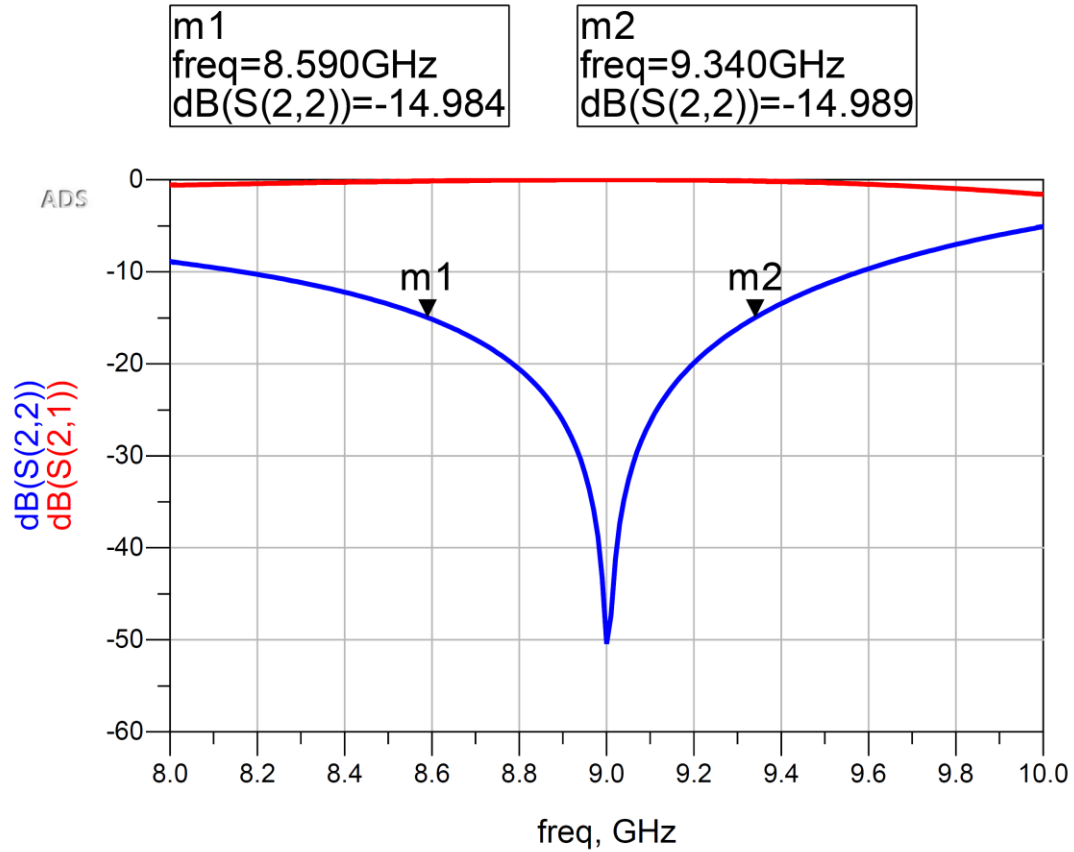


Figure 14 – Output Matching Network Design Verification

The output matching network design, when implemented on a Roger 3003 substrate (10 mil thickness), has an insertion loss of $S_{21} = -0.178\text{dB}$ and a return loss of $S_{22} = -20.541\text{dB}$ as shown in Figure 15. The insertion loss of -0.178dB is small and can be neglected. This justifies the assumption that the matching network is lossless.

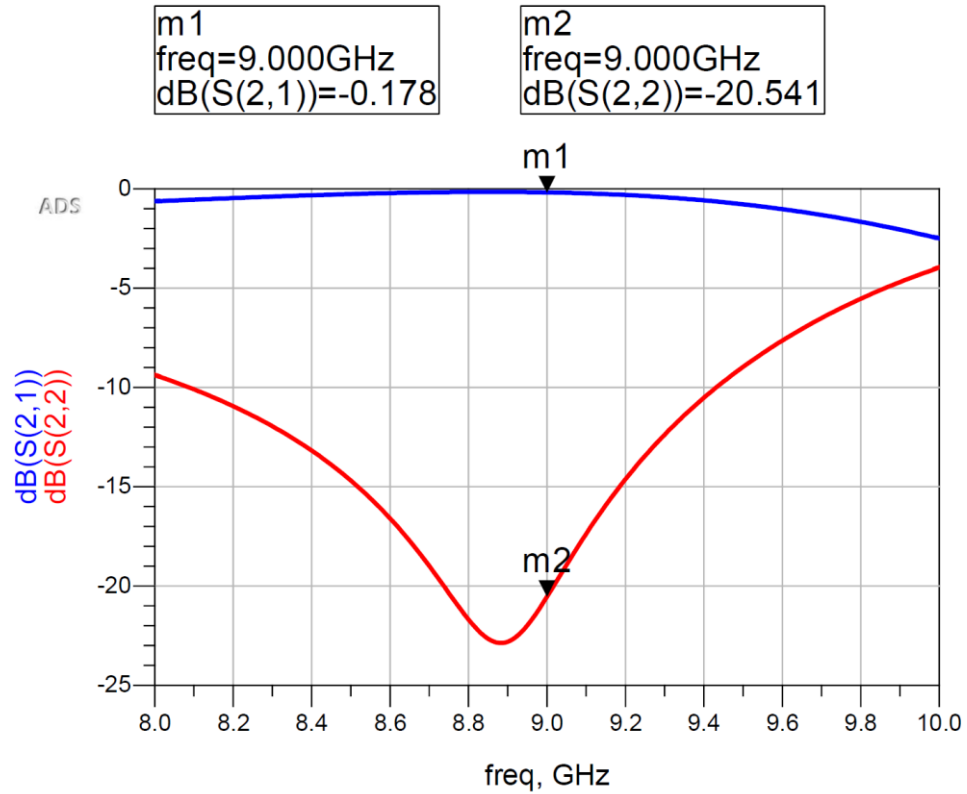


Figure 15 – Output Matching Network Design Performance

4.8 Input Matching Network

The input matching network is designed to transform the input impedance

$Z_{\text{IN}} = 1.973 + j8.212 \Omega$ ($\Gamma_{\text{IN}} = -0.877 + j0.297$) to 50Ω transmission line (conjugated matched for maximum power transferred). The matching network is a π -network consisting of two open shunt stubs and a series stub. The design is done in ADS Smith Chart Utility as shown below.

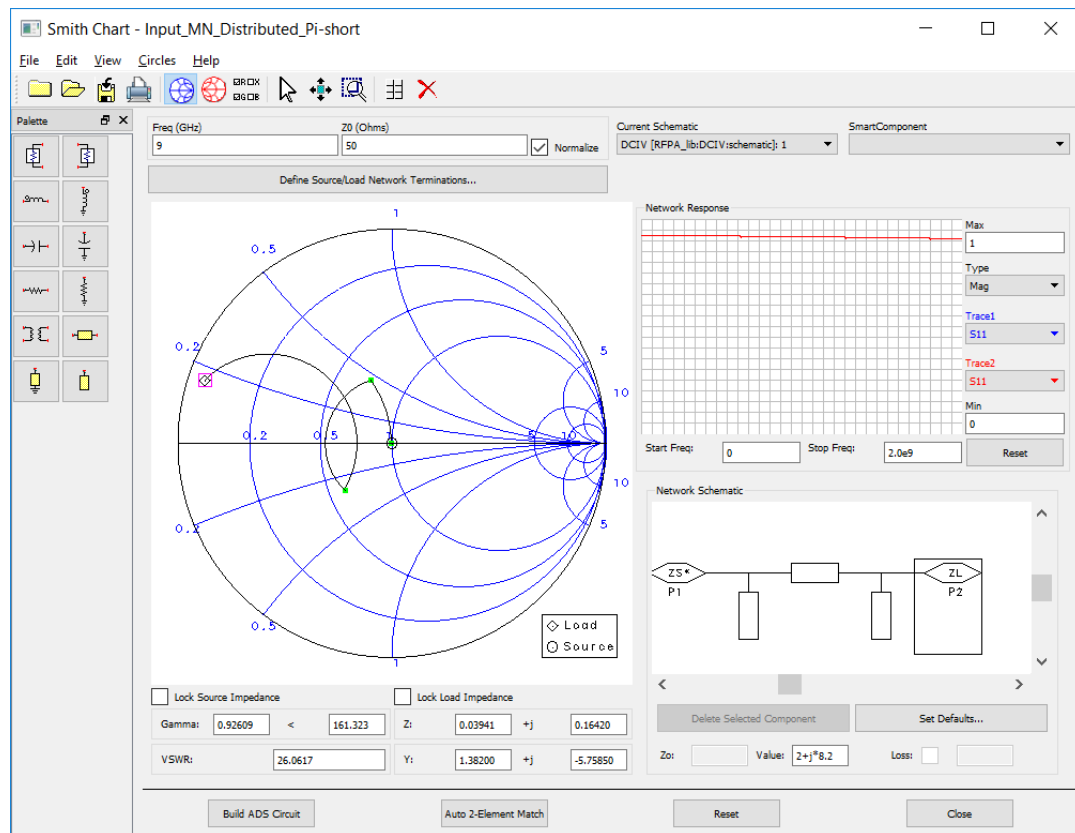


Figure 16 – Input Matching Network Design

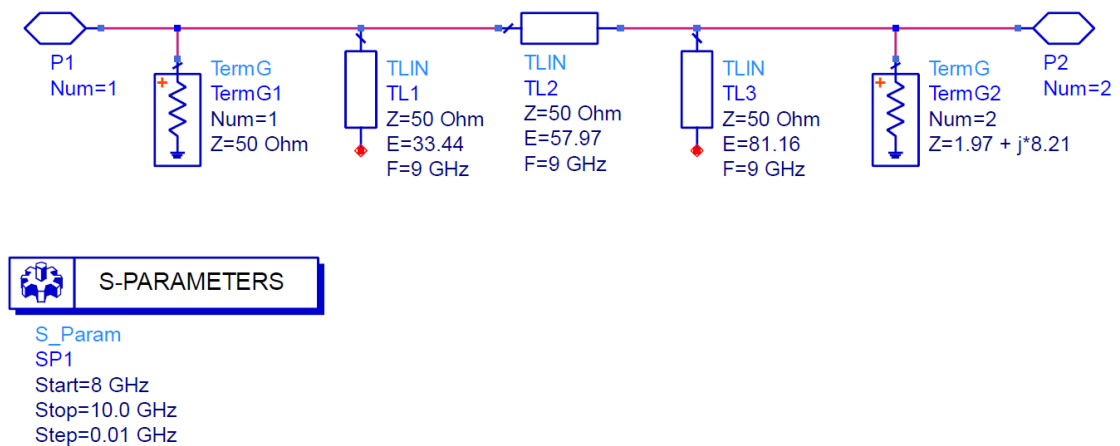


Figure 17 – Input Matching Network Design Schematic

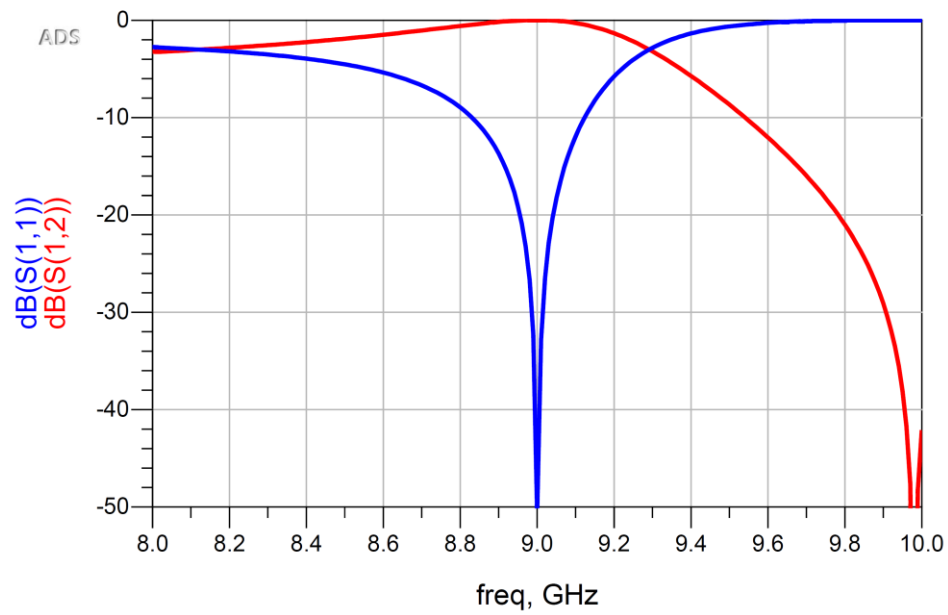


Figure 18 – Input Matching Network Verification

The input matching network design, when implemented on a Roger 3003 substrate (10 mil thickness), has an insertion loss of $S_{21} = -0.565\text{dB}$ and a return loss of $S_{11} = -31.46\text{dB}$ as shown in Figure 19. The insertion loss of -0.565dB is small and can be neglected.

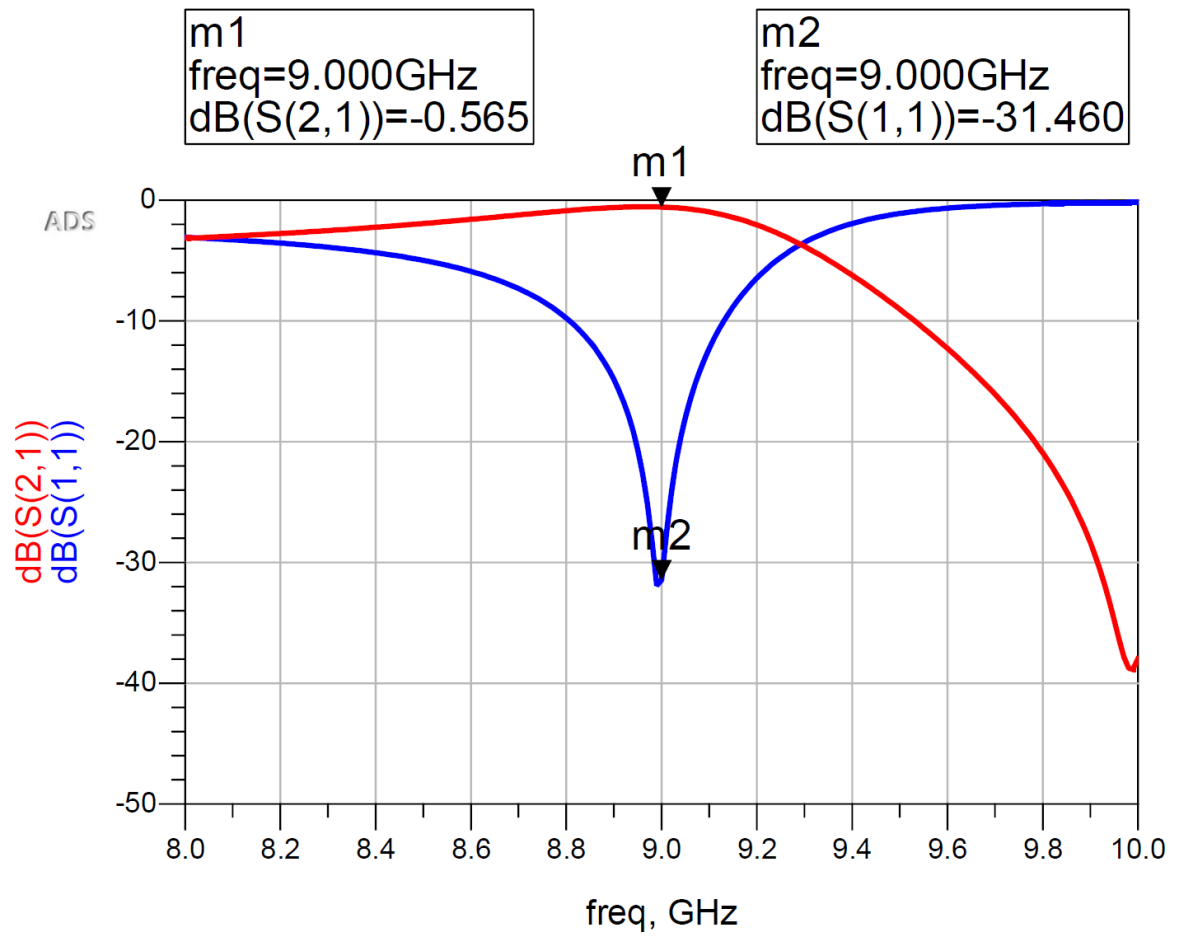


Figure 19 – Input Matching Network Design Performance

4.9 Complete Transistor Amplifier Design

After input and output matching networks are designed, the complete design is put together including the DC bias network. The bias network is consisted of an RFC, a shunt capacitor and quarter-wave transformer for isolation.

Overall amplifier performance is shown in Figures 24 - 27. The amplifier with input power of 28dBm has the output at 39.8dBm or over 11.8dBm gain. The PAE of the amplifier at Pavs of 28dBm is at 63.85%.

4.9.1 DC Block Capacitors

Input and output DC block capacitors are Interdigital capacitors. Its structure and design equations are in the Appendix E for reference. The capacitors are designed to have capacitance of 30pF or 0.6Ω at 9GHz.

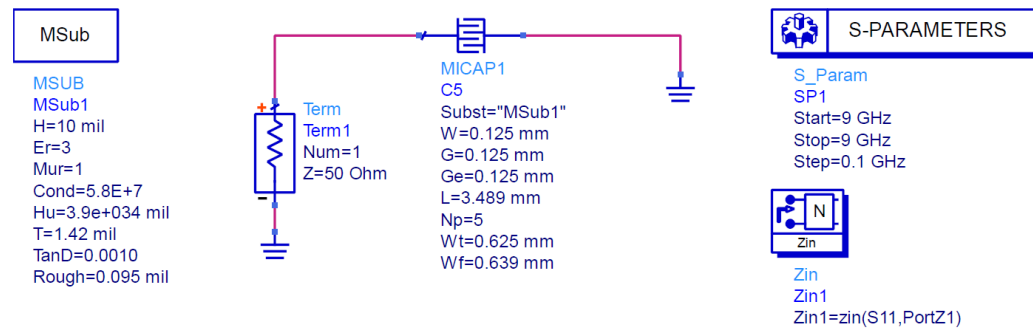


Figure 20 – Interdigital Capacitor Design

freq	Zin1	Zin1	C	Zc
9.000 GHz	0.058 - j0.585	0.588 / -84.330	30.239	0.585

$$\text{Eqn } C = -1/(2*\pi*freq*im(Zin1))*1e12$$

$$\text{Eqn } Zc = 1/(2*\pi*freq*C)*1e12$$

4.9.2 DC Bias Circuits

The function of DC bias circuit is to provide a low impedance path for DC voltage to the transistor but a high impedance path for RF from the transistor. This is necessary to provide proper bias of the transistor without loading the transistor and polluting DC source.

The DC bias circuit consists of an LC low-pass filter and a quarter-wave transformer. The LC (L=2.7nH and C=2pF) low-pass filter with a cut-off frequency $f_C=2.17\text{GHz}$ is used to prevent out-of-band RF frequencies from reaching the transistor. The component self-resonant frequencies of both L and C are above the operating frequency of 9GHz. A high impedance ($Z_{\lambda/4}=110\Omega$) quarter-wavelength transmission line is used to transform its low load impedance ($C_L=2\text{pF}$ or $Z_L=8.85\Omega$ at 9GHz) to a high input impedance Z_{IN} of:

$$Z_{IN} = \frac{\sqrt{Z_{\lambda/4}^2}}{Z_L} = 1.367k\Omega$$

This quarter-wave transmission line input impedance $Z_{IN}=1.367k\Omega$ ($\gg 50\Omega$) presents a high impedance to the RF signals at the drain of the transistor.

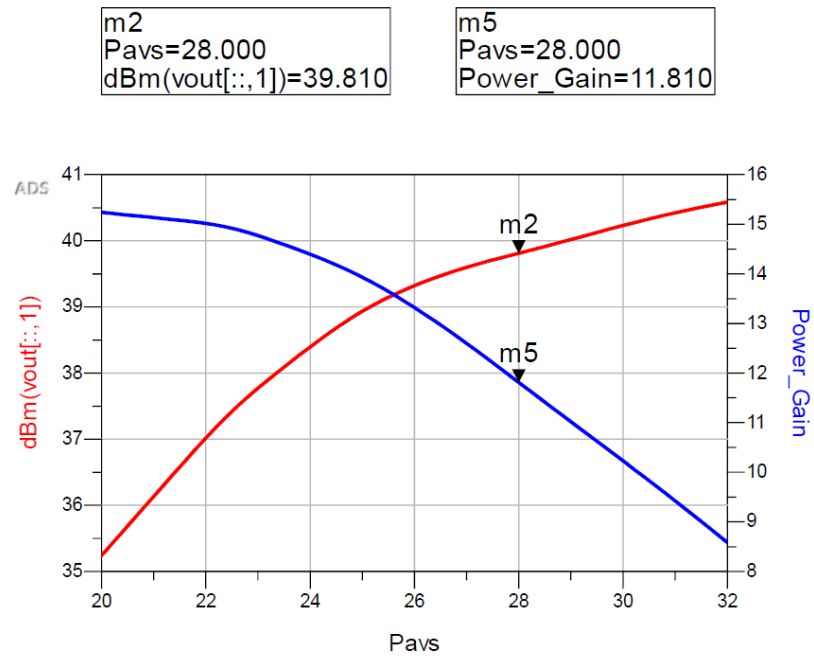


Figure 23 – Output Power and Power Gain vs Input Power P_{IN}

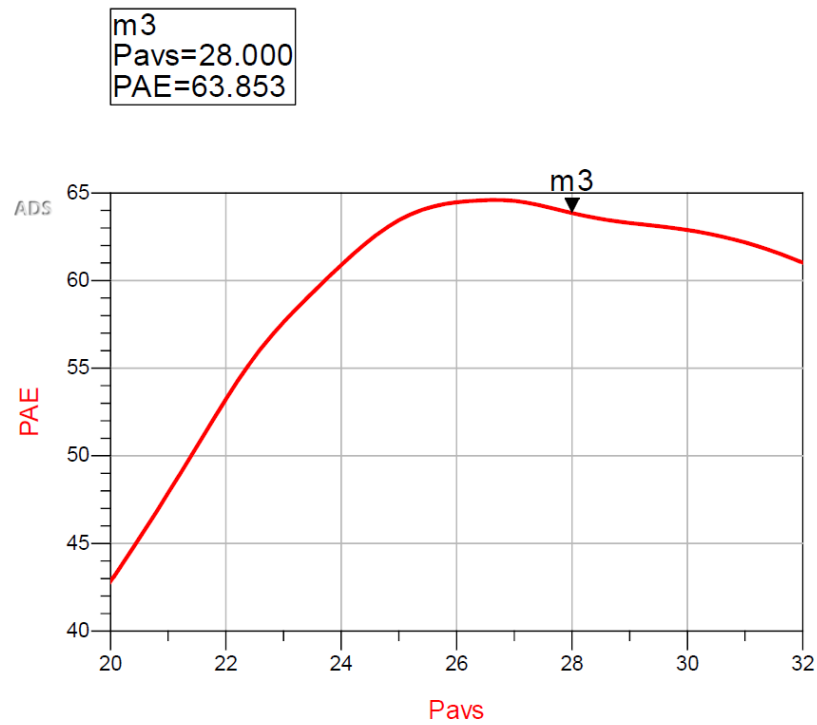


Figure 24 – PAE vs P_{IN}

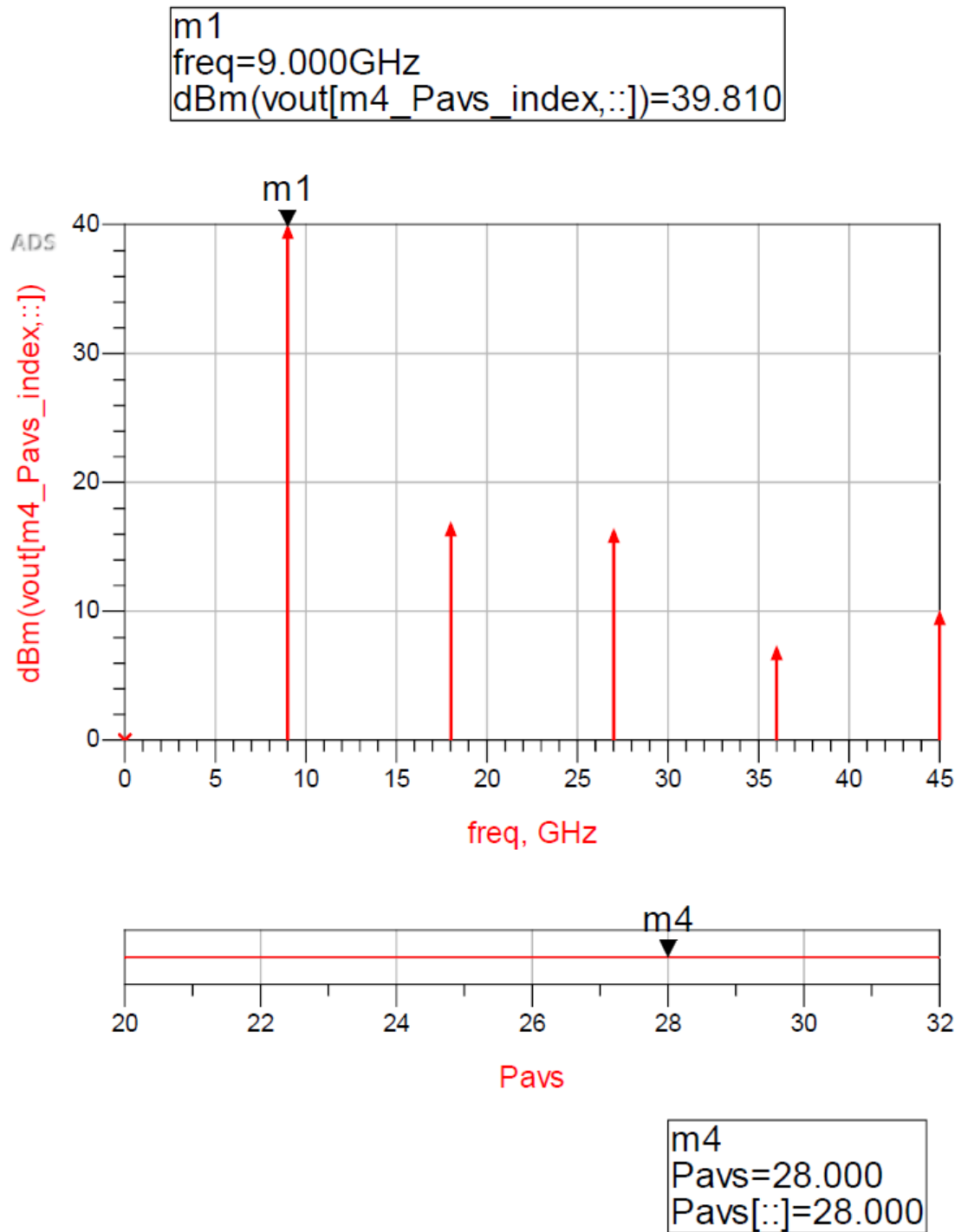


Figure 25 – Output Power Spectrum

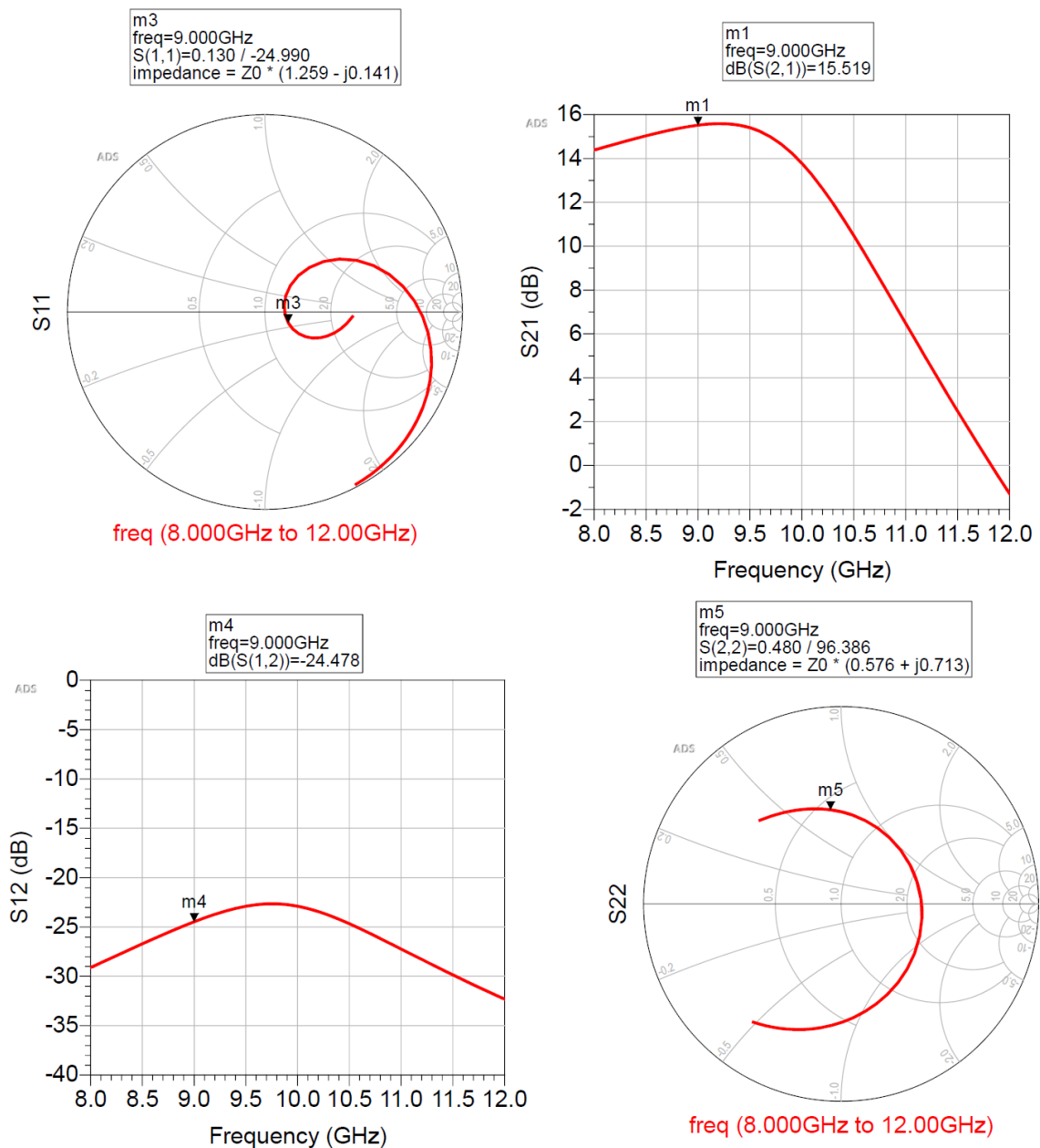


Figure 26 – Amplifier Small-Signal S-Parameters

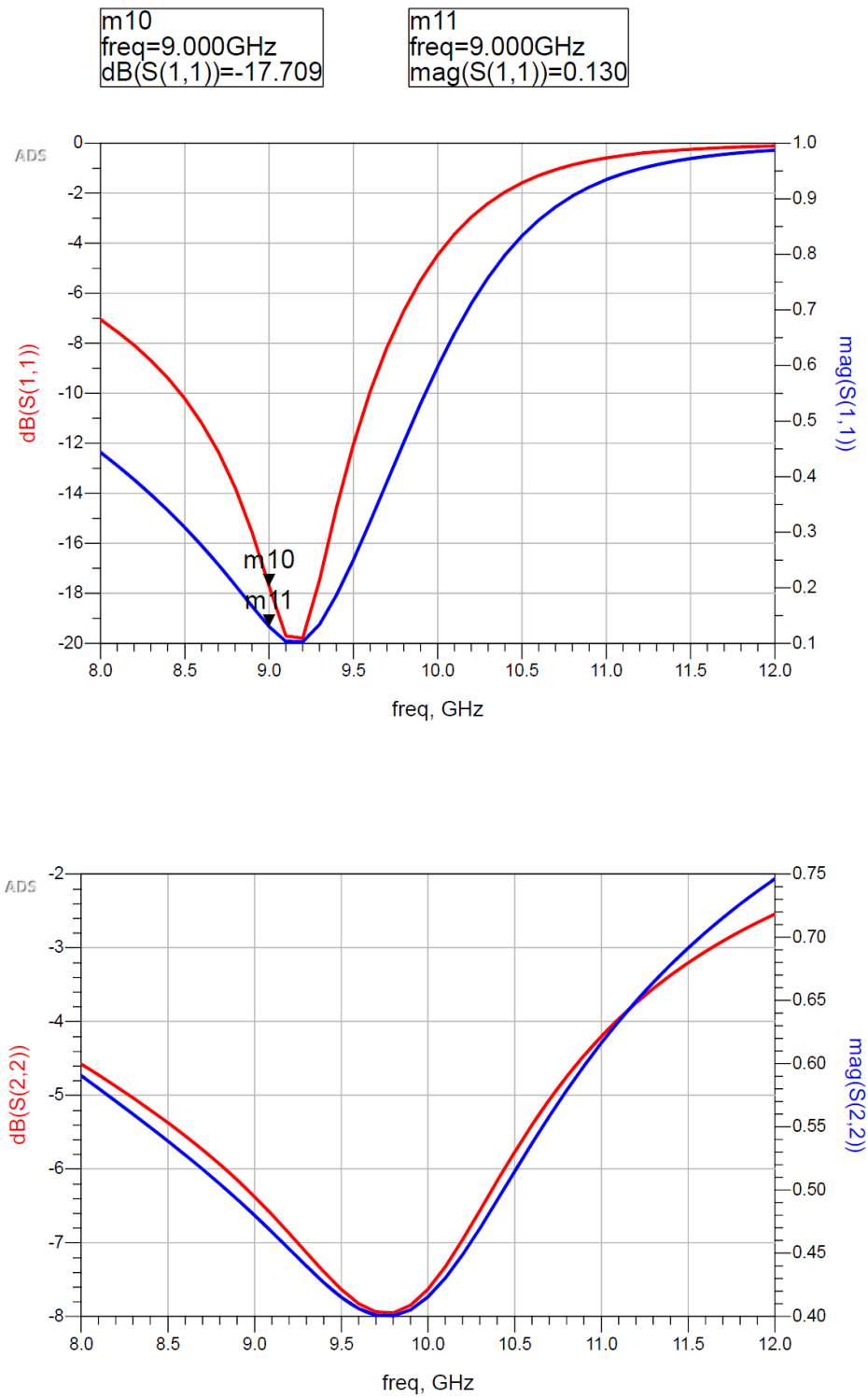


Figure 27 – Amplifier S₁₁ and S₂₂

Conclusion

In this project a single-stage power amplifier is designed using power contours-plot method, one of the three available design methods. Load reflection coefficient is selected from the maximum power contour to achieve maximum output power delivered to load. After Γ_L is selected, input reflection coefficient is calculated and used to find the source reflection coefficient. Once both Γ_{IN} and Γ_L are determined, input/output matching networks are designed to realize the desired input/output reflection coefficients. Matching networks are designed using lossless transmission lines in either L- or π -configuration. These simple matching networks are easy to manufacture but only perfectly matched at one frequency. Matching is degrading as frequency moves away from the intended frequency. Thus, the amplifiers designed with this kind of matching networks are narrow-band amplifiers. There are advanced techniques that can be used to design wide-band matching networks and are beyond the scope of this project.

Based on simulation results this design has met and has exceeded the design specifications. Output power is at 39.8dBm with a gain of 11.8dB and PAE of 63.8%. In this design, simulation is performed using ADS and Modelithics transistor non-linear model. The design could have been fabricated in CSUN and the amplifier's small-signal performance could have been tested using a Vector Network Analyzer (VNA). To verify the amplifier large-signal parameters (such as output power and gain, efficiency, linearity and distortion) a high power (>30dBm output) signal generator, a spectrum analyzer and an attenuator, and a power meter were needed. Future work will be to build a complete amplifier design, bench-test the design and compare actual measurement with simulation result.

Parameter	Goal	Simulation
Frequency	9GHz	9GHz
Output Power	39dBm	39.4dBm
Power Gain	10dB	11.4dB

Table 2 – Design Summary

References

- [1] Radmanesh, Matthew M., “Advanced RF & Microwave Circuit Design – The Ultimate Guide to Superior Design”, Bloomington, IN 47403, AuthorHouse 2018.
- [2] Radmanesh, Matthew M., “RF & Microwave Design Essentials – Engineering Design and Analysis from DC to Microwaves”, Bloomington, IN 47403, AuthorHouse 2007.
- [3] Radmanesh, Matthew M., “Radio Frequency and Microwave Electronics Illustrated”, Upper Saddle River, NJ 07458, Prentice Hall 2001.
- [4] Pozar, David M., “Microwave Engineering”, Fourth Edition, Hoboken NJ 07030, John Wiley & Sons, Inc. 2012.
- [5] Gonzalez, Guillermo, “Microwave Transistor Amplifiers – Analysis and Design”, Second Edition, Upper Saddle River, NJ 07458, Prentice Hall 1997.
- [6] Yeom, Kyung-Whan, “Microwave Circuit Design – A Practical Approach Using ADS”, Upper Saddle River, NJ 07458, Prentice Hall 2015.

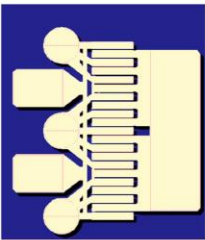
Appendix A – Qorvo TGF2023-2-02 Datasheet



TGF2023-2-02
12 Watt Discrete Power GaN on SiC HEMT

Applications

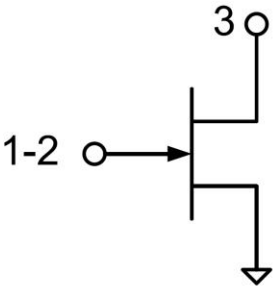
- Defense & Aerospace
- Broadband Wireless



Product Features

- Frequency Range: DC - 18 GHz
- 41.2 dBm Nominal P_{SAT} at 6 GHz
- 63.4% Maximum PAE at 6 GHz
- 18 dB Linear Gain at 6 GHz
- Bias: V_D = 12 - 32 V, I_{DQ} = 50 - 125 mA
- Technology: QGaN25 on SiC
- Chip Dimensions: 0.82 x 0.92 x 0.10 mm

Functional Block Diagram



General Description

The Qorvo TGF2023-2-02 is a discrete 2.5 mm GaN on SiC HEMT which operates from DC-18 GHz. The TGF2023-2-02 is designed using Qorvo's proven QGaN25 production process. This process features advanced field plate techniques to optimize microwave power and efficiency at high drain bias operating conditions.

The TGF2023-2-02 typically provides 41.2 dBm of saturated output power with power gain of 14.9 dB at 6GHz. The maximum power added efficiency is 63.4 % which makes the TGF2023-2-02 appropriate for high efficiency applications.

Lead-free and RoHS compliant

Pad Configuration

Pad No.	Symbol
1-2	V _G / RF IN
3	V _D / RF OUT
Backside	Source / Ground

Ordering Information

Part	ECCN	Description
TGF2023-2-02	EAR99	12 Watt GaN HEMT



TGF2023-2-02

12 Watt Discrete Power GaN on SiC HEMT

Absolute Maximum Ratings

Parameter	Value
Drain to Gate Voltage (V_{DG})	100 V
Gate Voltage Range (V_G)	-10 to 0 V
Drain Current (I_D)	2.5 A
Gate Current (I_G)	-2.5 to 7 mA
Power Dissipation (P_D)	See graph on pg.5.
CW Input Power (P_{IN})	+34 dBm
Channel Temperature (T_{CH})	275 °C
Mounting Temperature (30 Sec.)	320 °C
Storage Temperature	-65 to 150 °C

Operation of this device outside the parameter ranges given above may cause permanent damage. These are stress ratings only, and functional operation of the device at these conditions is not implied.

Recommended Operating Conditions⁽¹⁾

Parameter	Value
Drain Voltage Range (V_D)	12 – 40 V
Drain Quiescent Current (I_{DQ})	125 mA
Drain Current Under RF Drive (I_D)	750 mA (Typ.)
Gate Voltage (V_G)	-3.0 V (Typ.)
Channel Temperature (T_{CH})	225 °C (Max.)
Dissipation Power, CW (P_D) ³	8.2 W
Dissipation Power, Pulsed (P_D) ^{2,3}	11 W

1. Electrical specifications are measured at specified test conditions. Specifications are not guaranteed over all recommended operating conditions.

2. 684 uS Pulse Width, 10% Duty Cycle

3. Carrier plate temperature is at 85°C.

RF Characterization – Optimum Power Tune

Test conditions unless otherwise noted: T = 25 °C.

Parameter	Typical Value								Units
Frequency (F)	3		6		8		10		GHz
Drain Voltage (V_D)	28	28	28	28	28	28	28	28	V
Bias Current (I_{DQ})	50	125	50	125	50	125	50	125	mA
Output P3dB (P_{3dB})	41.1	41	41.3	41.2	41.3	41.2	41.3	41.2	dBm
PAE @ P3dB (PAE_{3dB})	60.9	60.1	59.1	59.0	57.2	57.0	55	55.4	%
Gain @ P3dB (G_{3dB})	19.4	20.3	14.1	14.9	11.8	12.5	10.0	10.8	dB
Parallel Resistance ⁽¹⁾ (R_p)	63.7	63.6	61.3	62.1	58.7	58.3	54.7	54.4	Ω-mm
Parallel Capacitance ⁽¹⁾ (C_p)	0.287	0.275	0.315	0.313	0.316	0.317	0.328	0.329	pF/mm
Load Reflection Coefficient ⁽²⁾ (Γ_L)	0.445±18°	0.44±17°	0.46±39°	0.46±38°	0.47±50°	0.47±50°	0.48±63°	0.48±63°	--

Notes:

1. Large signal equivalent output network (normalized).

2. Characteristic Impedance (Z_0) = 10 Ω.

RF Characterization – Optimum Efficiency Tune

Test conditions unless otherwise noted: T = 25 °C.

Parameter	Typical Value								Units
Frequency (F)	3		6		8		10		GHz
Drain Voltage (V_D)	28	28	28	28	28	28	28	28	V
Bias Current (I_{DQ})	50	125	50	125	50	125	50	125	mA
Output P3dB (P_{3dB})	39.7	39.8	40.1	40.2	40.1	40.2	40.4	40.3	dBm
PAE @ P3dB (PAE_{3dB})	66.2	65	64.2	63.4	62.2	61.8	59.5	59.7	%
Gain @ P3dB (G_{3dB})	20.9	21.7	15.4	16.1	13.2	13.5	11.1	11.7	dB
Parallel Resistance ⁽¹⁾ (R_p)	111	108	101	97.1	96.4	91.6	83.4	83	Ω-mm
Parallel Capacitance ⁽¹⁾ (C_p)	0.357	0.350	0.368	0.365	0.376	0.376	0.373	0.381	pF/mm
Load Reflection Coefficient ⁽²⁾ (Γ_L)	0.64±20°	0.63±20°	0.64±40°	0.63±40°	0.65±53°	0.64±53°	0.64±64°	0.64±65°	--

Notes:

1. Large signal equivalent output network (normalized).

2. Characteristic Impedance (Z_0) = 10 Ω.



Thermal and Reliability - CW ⁽¹⁾

Parameter	Test Conditions	Value	Units
Thermal Resistance, θ_{JC}	$P_D = 2.5 \text{ W}$, $T_{\text{baseplate}} = 85^\circ\text{C}$	12.0	$^\circ\text{C/W}$
Channel Temperature, T_{CH}		115	$^\circ\text{C}$
Median Lifetime, T_M		1.9E11	Hrs
Thermal Resistance, θ_{JC}	$P_D = 5.0 \text{ W}$, $T_{\text{baseplate}} = 85^\circ\text{C}$	13.0	$^\circ\text{C/W}$
Channel Temperature, T_{CH}		150	$^\circ\text{C}$
Median Lifetime, T_M		2.4E9	Hrs
Thermal Resistance, θ_{JC}	$P_D = 7.5 \text{ W}$, $T_{\text{baseplate}} = 85^\circ\text{C}$	14.0	$^\circ\text{C/W}$
Channel Temperature, T_{CH}		190	$^\circ\text{C}$
Median Lifetime, T_M		4.0E7	Hrs
Thermal Resistance, θ_{JC}	$P_D = 10.0 \text{ W}$, $T_{\text{baseplate}} = 85^\circ\text{C}$	15.4	$^\circ\text{C/W}$
Channel Temperature, T_{CH}		239	$^\circ\text{C}$
Median Lifetime, T_M		6.1E5	Hrs
Thermal Resistance, θ_{JC}	$P_D = 12.5 \text{ W}$, $T_{\text{baseplate}} = 85^\circ\text{C}$	17.1	$^\circ\text{C/W}$
Channel Temperature, T_{CH}		299	$^\circ\text{C}$
Median Lifetime, T_M		9.0E3	Hrs

Notes:

1. Assumes eutectic attach using 1.5 mil thick 80/20 AuSn mounted to a 10 mm x 10 mm x 40 mil CuMo Carrier Plate.

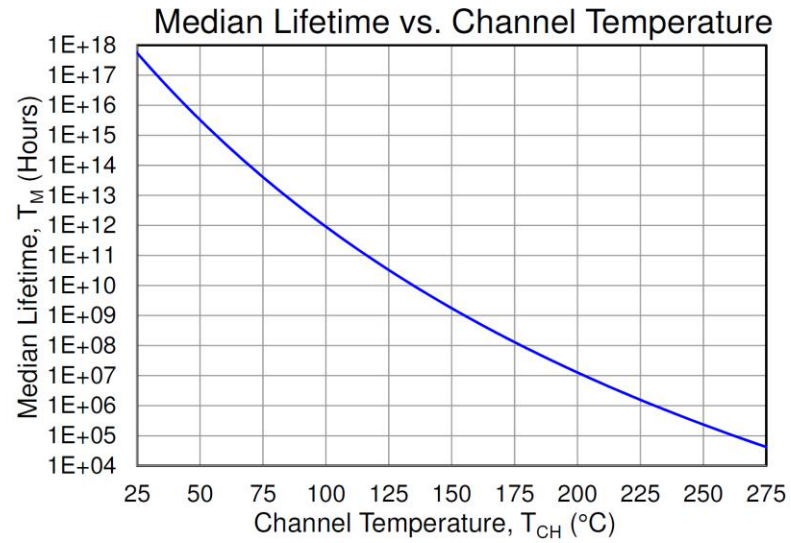
Thermal and Reliability - Pulsed ⁽¹⁾

Parameter	Test Conditions	Value	Units
Thermal Resistance, θ_{JC}	$P_D = 11 \text{ W}$, $T_{\text{baseplate}} = 85^\circ\text{C}$ Pulse Width = 100 μS Duty Cycle = 5%	9.5	$^\circ\text{C/W}$
Channel Temperature, T_{CH}		190	$^\circ\text{C}$
Median Lifetime, T_M		8.0E8	Hrs
Thermal Resistance, θ_{JC}	$P_D = 11 \text{ W}$, $T_{\text{baseplate}} = 85^\circ\text{C}$ Pulse Width = 100 μS Duty Cycle = 10%	9.8	$^\circ\text{C/W}$
Channel Temperature, T_{CH}		193	$^\circ\text{C}$
Median Lifetime, T_M		3.0E8	Hrs
Thermal Resistance, θ_{JC}	$P_D = 11 \text{ W}$, $T_{\text{baseplate}} = 85^\circ\text{C}$ Pulse Width = 100 μS Duty Cycle = 20%	10.4	$^\circ\text{C/W}$
Channel Temperature, T_{CH}		199	$^\circ\text{C}$
Median Lifetime, T_M		8.5E7	Hrs
Thermal Resistance, θ_{JC}	$P_D = 11 \text{ W}$, $T_{\text{baseplate}} = 85^\circ\text{C}$ Pulse Width = 100 μS Duty Cycle = 50%	12.4	$^\circ\text{C/W}$
Channel Temperature, T_{CH}		221	$^\circ\text{C}$
Median Lifetime, T_M		5.1E6	Hrs

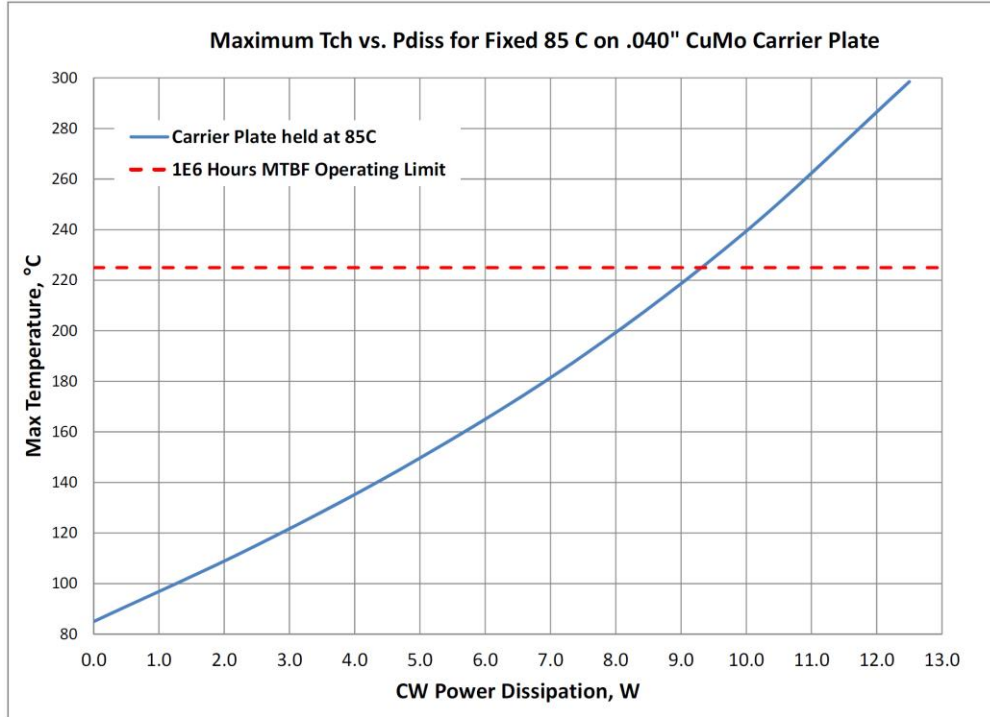
Notes:

1. Assumes eutectic attach using 1.5 mil thick 80/20 AuSn mounted to a 10 mm x 10 mm x 40 mil CuMo Carrier Plate.

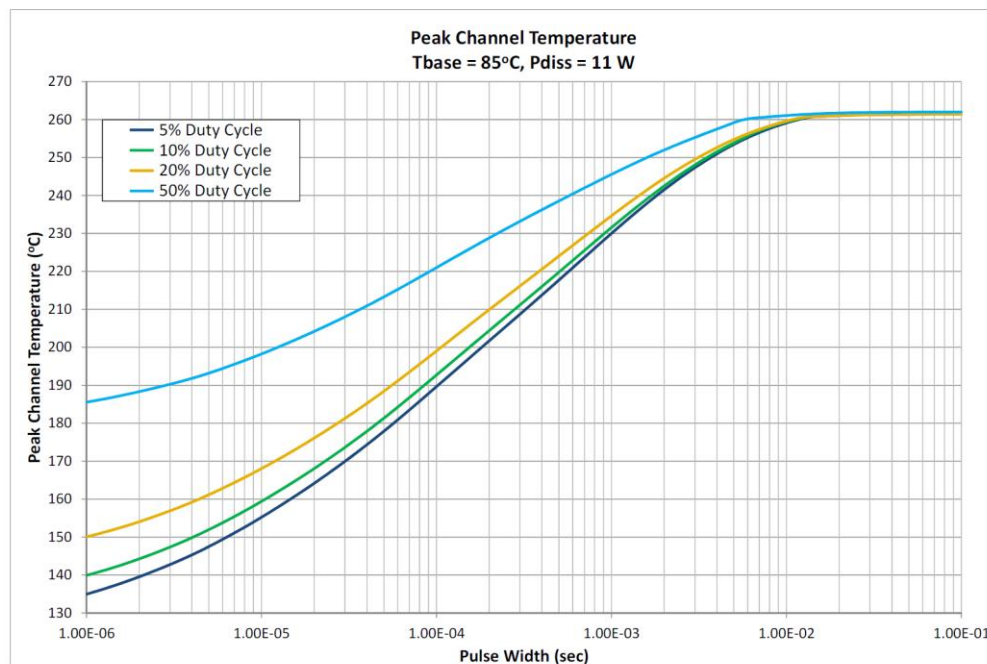
Median Lifetime



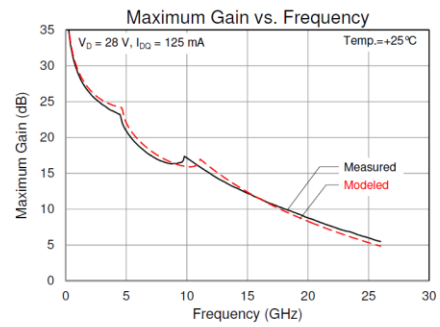
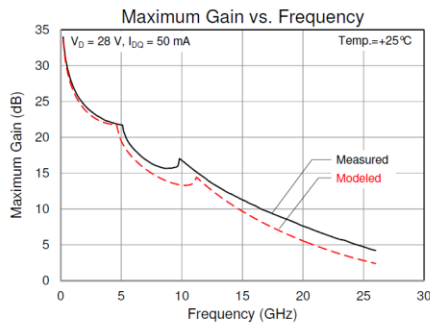
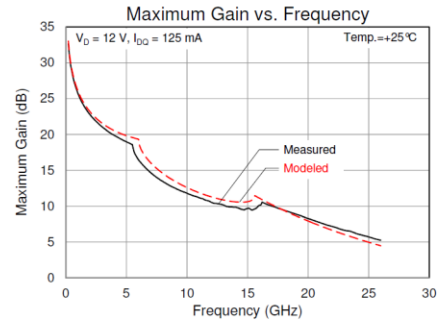
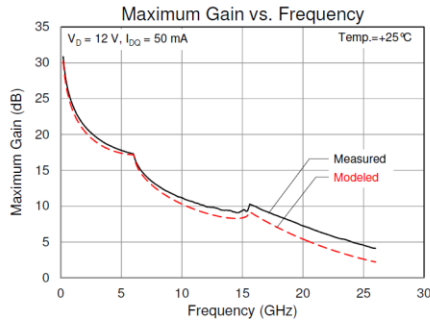
Maximum Channel Temperature - CW



Peak Channel Temperature - Pulsed

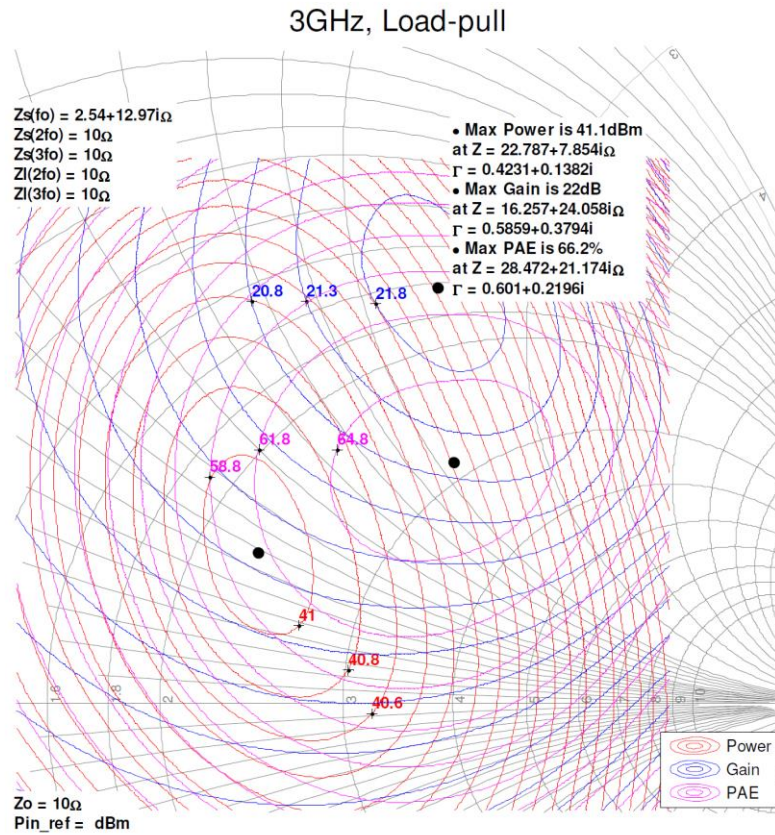


Measured Maximum Gain



Model Load Pull Contours

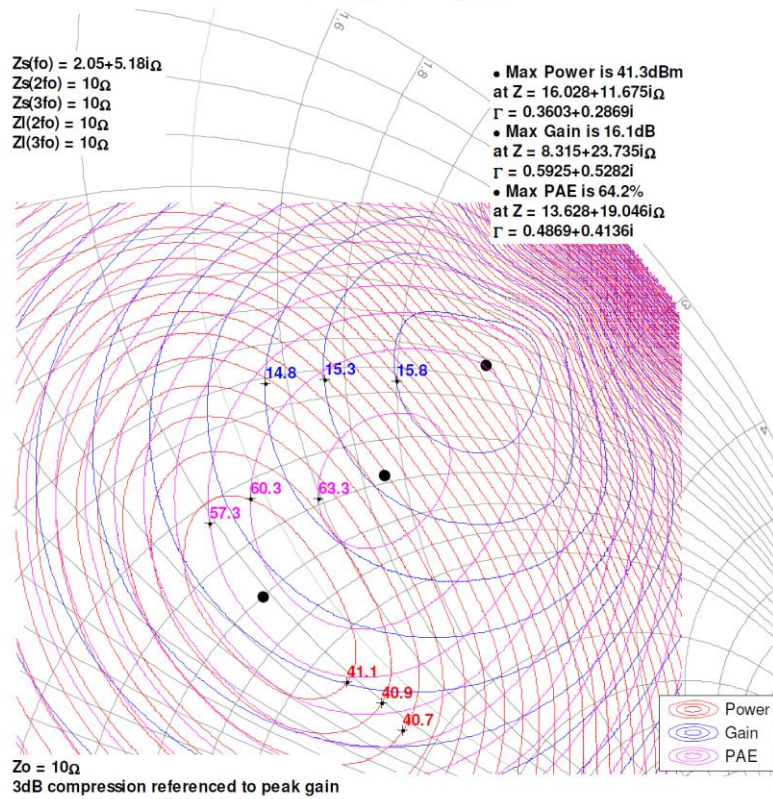
Simulated signal: 10% pulses
Vd = 28 V, Idq = 50 mA



Model Load Pull Contours

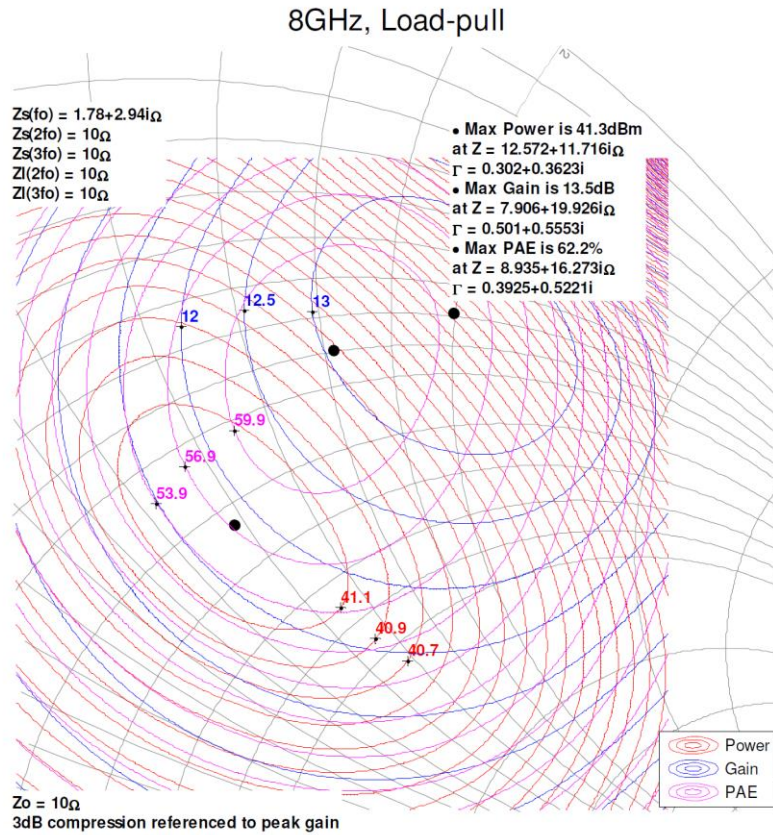
Simulated signal: 10% pulses
Vd = 28 V, Idq = 50 mA

6GHz, Load-pull



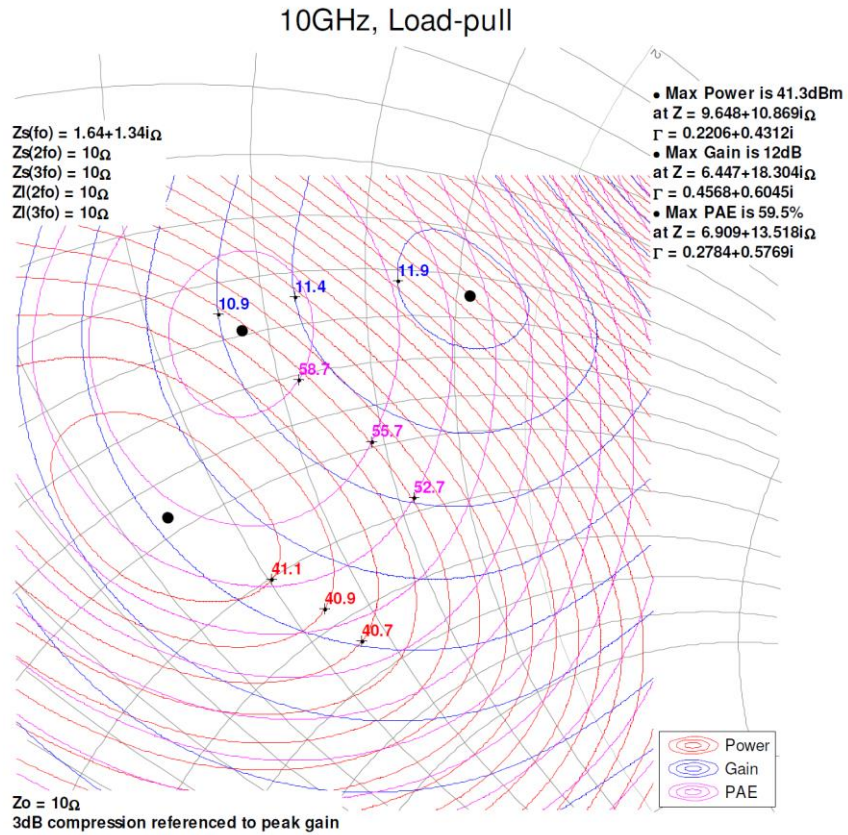
Model Load Pull Contours

Simulated signal: 10% pulses
Vd = 28 V, Idq = 50 mA



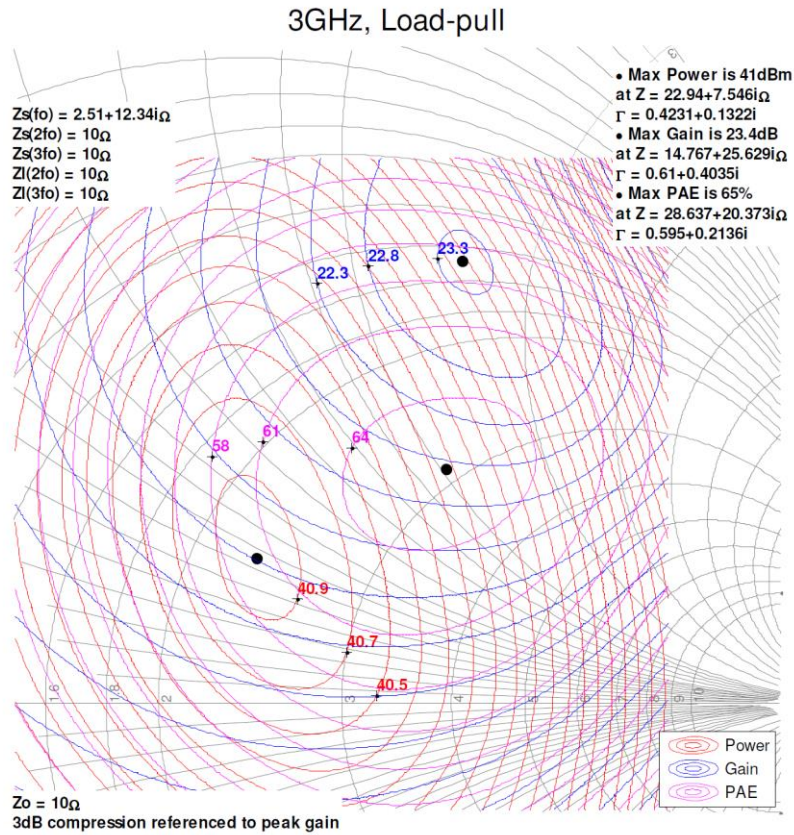
Load Pull Contours

Simulated signal: 10% pulses
Vd = 28 V, Idq = 50 mA



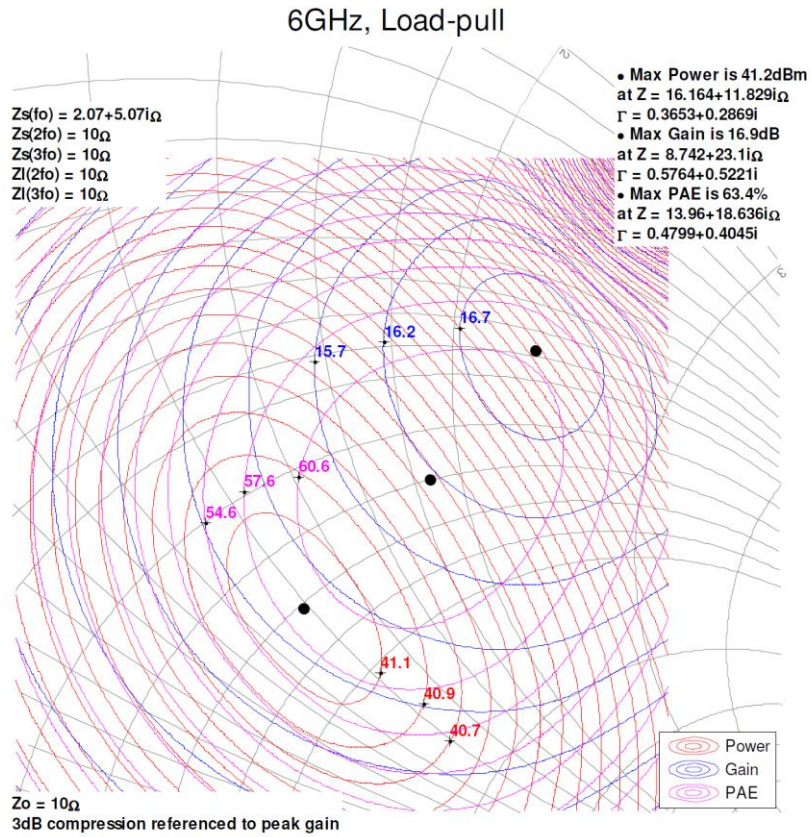
Load Pull Contours

Simulated signal: 10% pulses
Vd = 28 V, Idq = 125 mA



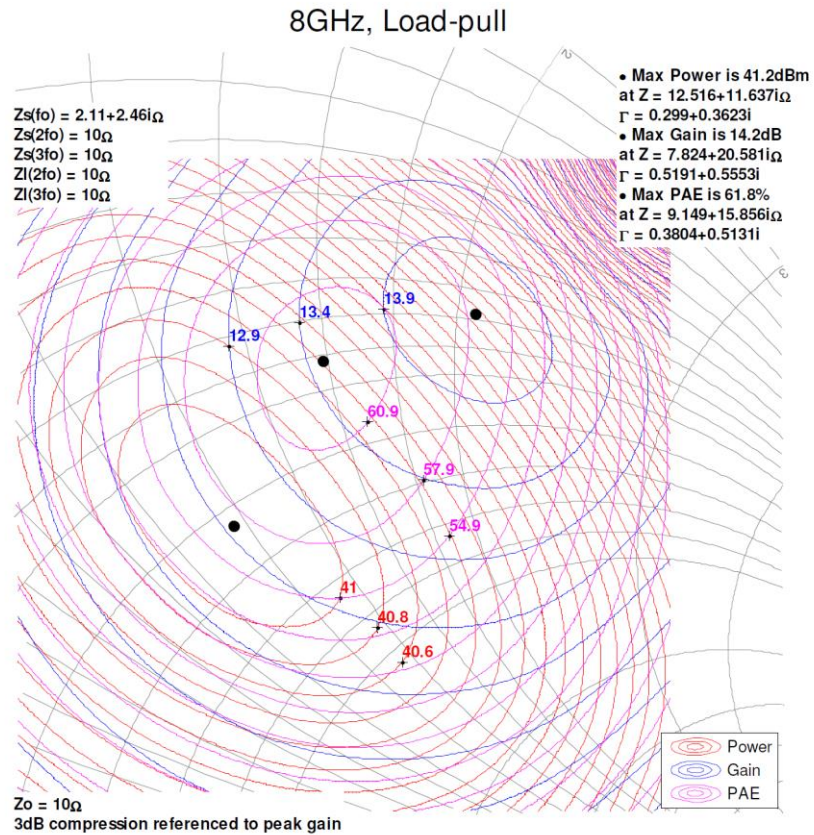
Load Pull Contours

Simulated signal: 10% pulses
Vd = 28 V, Idq = 125 mA



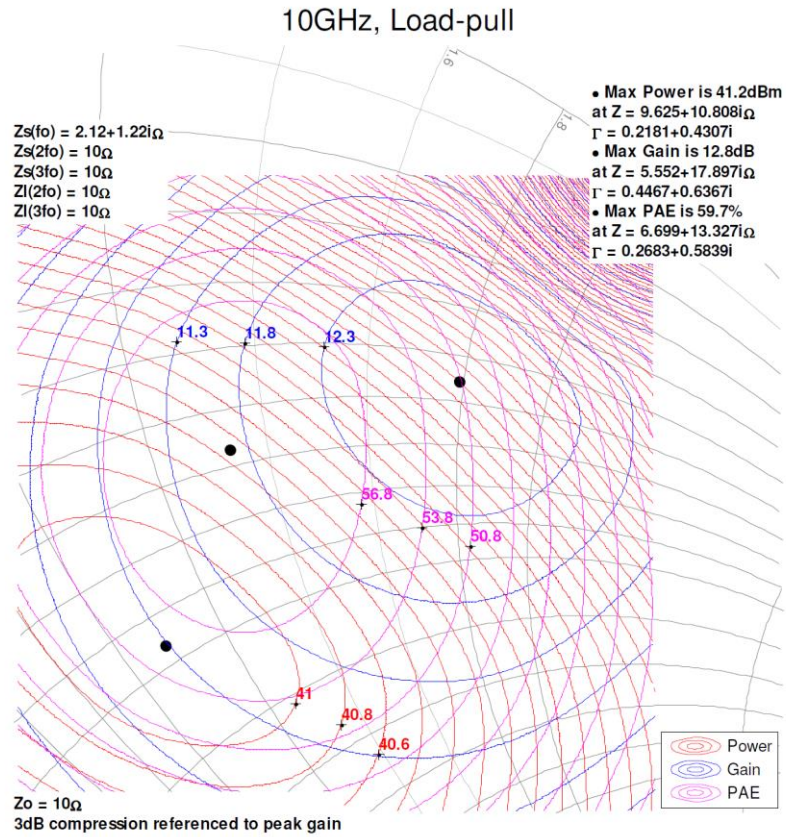
Load Pull Contours

Simulated signal: 10% pulses
Vd = 28 V, Idq = 125 mA

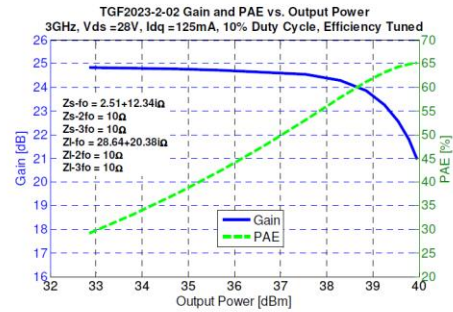
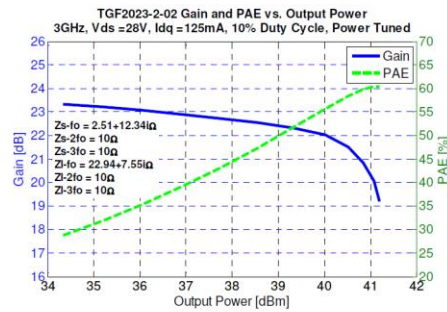
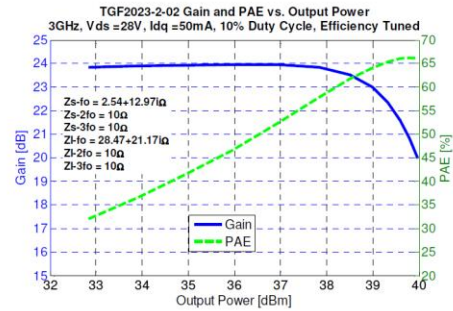
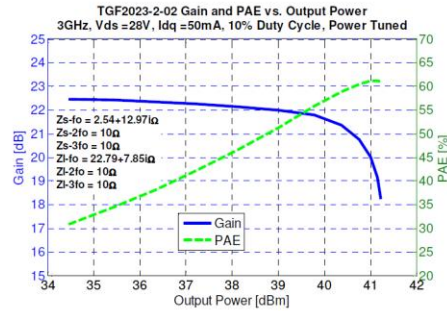


Load Pull Contours

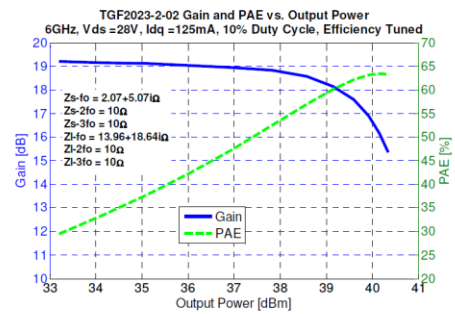
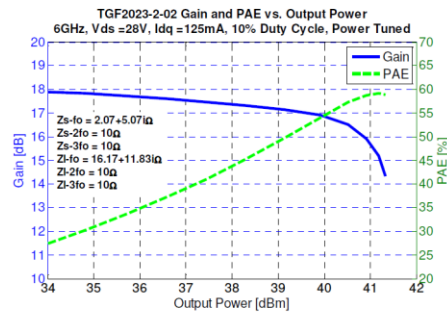
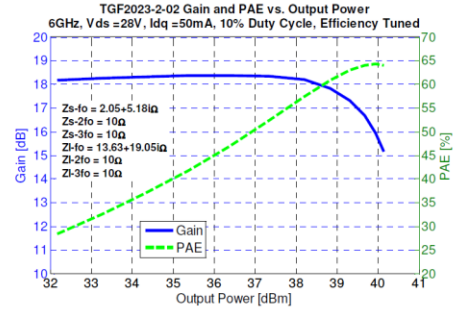
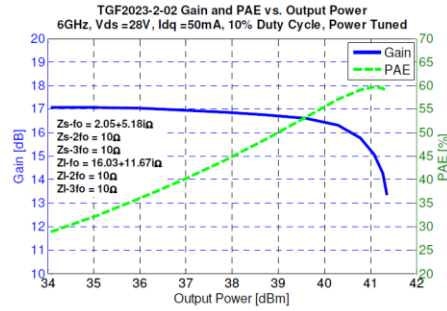
Simulated signal: 10% pulses
Vd = 28 V, Idq = 125 mA



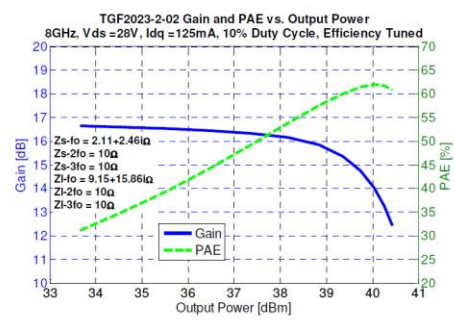
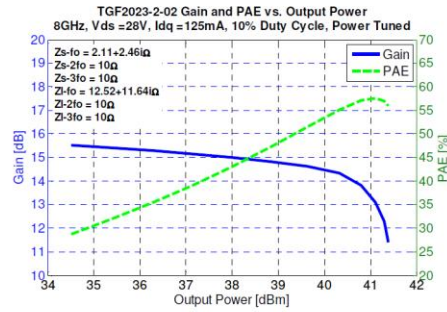
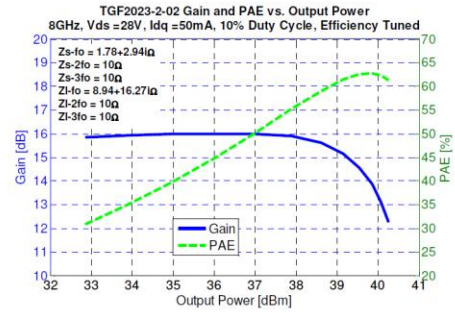
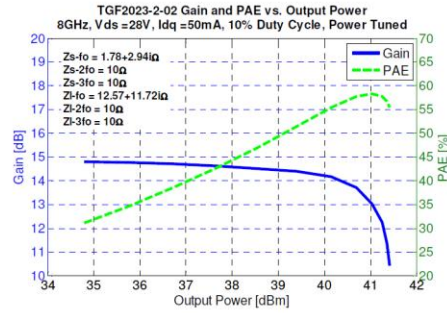
Model Drive-up Data – 3 GHz



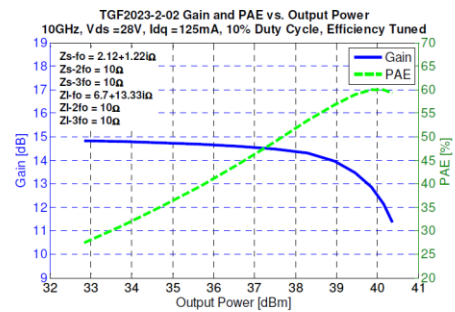
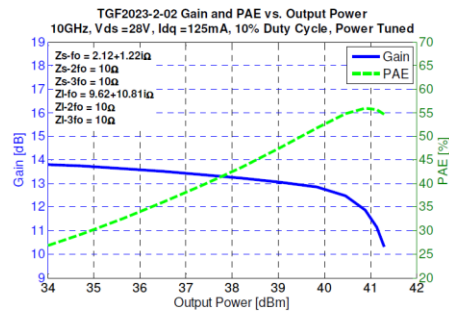
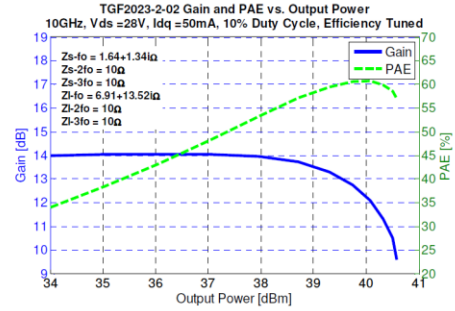
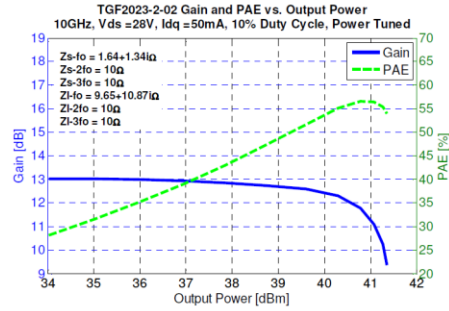
Model Drive-up Data – 6 GHz



Model Drive-up Data – 8 GHz



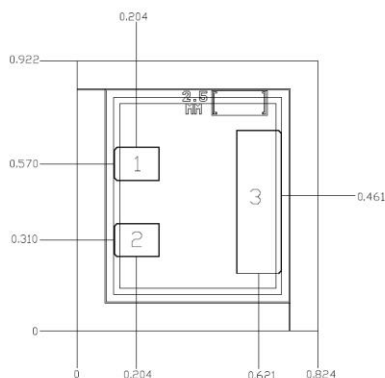
Model Drive-up Data – 10 GHz



Model

A non-linear model is available for download from Modelithics (at <http://www.modelithics.com/mvp/Qorvo&tab=3>) by approved Qorvo customers. The model is compatible with the industry's most popular design software including Agilent ADS and National Instruments/AWR applications. Once on the Modelithics web page, the user will need to register for a free license before being granted the download.

Mechanical Drawing



Notes:

1. Units: millimeters
2. Thickness: 0.100 mm
3. Die x,y size tolerance: +/- 0.050 mm

Bond Pads

Pad No.	Description	Dimensions
1-2	Gate	0.154 x 0.115
3	Drain	0.154 x 0.490
Die Backside	Source / Ground	0.824 x 0.922

Assembly Notes

Component placement and adhesive attachment assembly notes:

- Vacuum pencils and/or vacuum collets are the preferred method of pick up.
- Air bridges must be avoided during placement.
- The force impact is critical during auto placement.
- Organic attachment (i.e. epoxy) not recommended.

Reflow process assembly notes:

- Use AuSn (80/20) solder and limit exposure to temperatures above 300°C to 3-4 minutes, maximum.
- An alloy station or conveyor furnace with reducing atmosphere should be used.
- Do not use any kind of flux.
- Coefficient of thermal expansion matching is critical for long-term reliability.
- Devices must be stored in a dry nitrogen atmosphere.

Interconnect process assembly notes:

- Ball bonding is the preferred interconnect technique, except where noted on the assembly diagram.
- Force, time, and ultrasonics are critical bonding parameters.
- Aluminum wire should not be used.
- Devices with small pad sizes should be bonded with 0.0007-inch wire.

Disclaimer

GaN/SiC devices are susceptible to damage from Electrostatic Discharge. Proper precautions should be observed during handling, assembly and test.

Bias-up Procedure

1. Set V_G to -5 V.
2. Set I_D limit to 140 mA.
3. Set V_D to 28 V.
4. Adjust V_G more positive until quiescent I_D is 125 mA.
5. Set I_D limit to 1 A.
6. Apply RF signal.

Bias-down Procedure

1. Turn off RF signal.
2. Turn off V_D and wait 1 second to allow drain capacitor dissipation.
3. Turn off V_G .

Product Compliance Information

ESD Sensitivity Ratings



Caution! ESD-Sensitive Device

GaN devices are susceptible to damage from Electrostatic Discharge. Proper precautions should be observed during handling, assembly and test.

Not HAST compliant.

Solderability

Compatible with gold/tin (320 °C maximum reflow temperature) soldering processes.

RoHS Compliance

This part is compliant with EU 2002/95/EC RoHS directive (Restrictions on the Use of Certain Hazardous Substances in Electrical and Electronic Equipment).

This product also has the following attributes:

- Lead Free
- Halogen Free (Chlorine, Bromine)
- Antimony Free
- TBBP-A (C₁₅H₁₂Br₄O₂) Free
- PFOS Free
- SVHC Free

Contact Information

For the latest specifications, additional product information, worldwide sales and distribution locations, and information about Qorvo:

Web: www.qorvo.com Tel: +1.972.994.8465
 Email: info-sales@qorvo.com Fax: +1.972.994.8504

For technical questions and application information: Email: info-products@qorvo.com

Important Notice

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Qorvo products are not warranted or authorized for use as critical components in medical, life-saving, or life-sustaining applications, or other applications where a failure would reasonably be expected to cause severe personal injury or death.

Appendix B – Modelithics Qorvo GaN Library



World's Best RF and Microwave
Simulation Models

Modelithics® Qorvo GaN Library

Overview

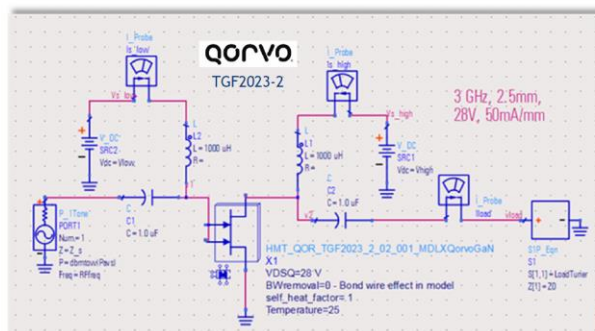
The Modelithics® Qorvo GaN Library is a collection of precision non-linear simulation models for Qorvo GaN die and packaged transistor devices. The GaN transistor models offer valuable features to allow for the most current and accurate design techniques, such as variable bias, temperature scaling, self-heating effects and intrinsic I-V sensing. The die transistor models offer bondwire on/off modes whenever applicable. Every model is validated by accurate broadband measurements.

Model Features

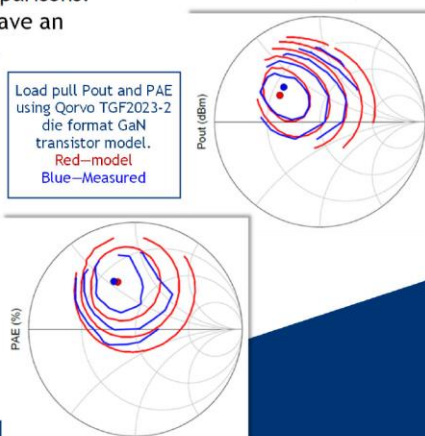
The Modelithics Qorvo GaN Library provides designers with a very flexible and accurate electronic design tool that offers many advantages over ideal or file-based models. Valuable features include:

- **Support for Popular Simulation Software***—Model versions are available for Keysight Advanced Design System and National Instruments AWR Design Environment
- **Measurement-based**—Each non-linear model is developed using multiple specialized measurements under device-specific test conditions.
- **Examples**—The library includes example design files that demonstrate the model features, various simulation test bench setups, and model performance.
- **Well-Documented**—Each model contains a model datasheet that lists recommended model validity parameters, measurement and test fixture details, and model-to-measurement data comparisons.
- **Power Amplifier Validations**—Select Qorvo GaN models have an associated PA design, including documented design steps.

*Models available may vary between simulators.



Load pull Pout and PAE using Qorvo TGF2023-2 die format GaN transistor model.
Red—model
Blue—Measured



Precision Measurements and Models You Trust

©Modelithics, Inc. 10/2017

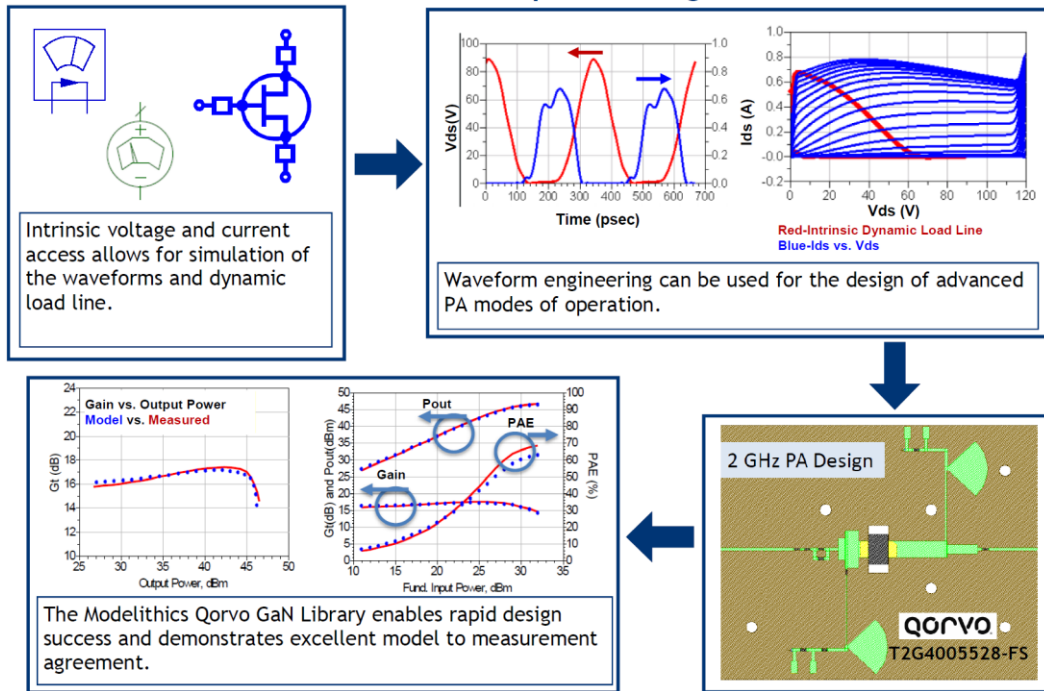
Devices in the Modelithics® Qorvo GaN Library

Packaged Devices				Die Devices	
TGF2819-FL	T1G3000532-SM	T2G6001528-SG	QPD1013	TGF2023-2-01	TGF2957
TGF2819-FS	T1G4004532-FL	T2G6003028-FL	QPD1014	TGF2023-2-02	TGF2933*
TGF2929-FL	T1G4004532-FS	T2G6003028-FS	QPD1015	TGF2023-2-05	TGF2934*
TGF2929-FS	T1G4012036-FL	TQP0104	QPD1015L	TGF2023-2-10	TGF2935*
TGF2965-SM	T1G4012036-FS	QPD0030	QPD1016	TGF2023-2-20	TGF2936*
TGF2977-SM	T1G4020036-FL	QPD0060	QPD1020	TGF2952	TGF2941*
TGF2978-SM	T1G4020036-FS	QPD1000	QPD1022	TGF2953	TGF2942*
TGF2979-SM	T1G6001032-SM	QPD1004	QPD1823	TGF2954	* Both small- and large-signal model versions
TGF3015-SM	T2G4003532-FL	QPD1008	QPD2194	TGF2955	
TGF3020-SM	T2G4003532-FS	QPD1008L	QPD2195	TGF2956	
TGF3021-SM	T2G4005528-FS	QPD1009	QPD2796		
T1G2028536-FL	T2G6000528-Q3	QPD1010	QPD3601		
T1G2028536-FS	T2G6001528-Q3	QPD1011	QPD3800		

V17.2.6

More to come! New models are added continually. [Visit our website for an updated complete list.](#)

Power Amplifier Design



Visit the Qorvo MVP page on the Modelithics website to:

- Explore the latest list of available Qorvo GaN models
- View model datasheets
- Access application notes that demonstrate the model features
- Request* the Modelithics Qorvo GaN Library at www.Modelithics.com/MVP/Qorvo *with approval

Precision Measurements and Models You Trust

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Appendix C – TGF2023-2-02_28V_125mA S-Parameters

!Data measured by Modelithics Inc. support@modelithics.com 813-866-6335 Tampa FL USA
 ! 2-Port S-parameters saved by WinCal
 ! VAR MeasName=S-Parameters (CALIBRATED_DATA) read from VNA (N5247A)
 ! VAR MeasDate=6/27/2013 3:18:33 PM
 ! VAR NAME=Measurement 4
 ! VAR FILENAME=2p5mm_task1_28p1Vds_m2p77Vgs_127mA_25C_S8.s2p
 ! VAR DATE=6/27/2013 3:18:33 PM
 ! VAR PHYS_PORTS=1,2
 !

Hz S R I R 50

2.00E+08	0.76428	-0.59386	-27.9892	11.5816	0.00277	0.0073	0.24943	-0.13761
3.00E+08	0.55237	-0.77666	-24.1716	14.82093	0.00529	0.00952	0.20407	-0.18839
4.00E+08	0.33441	-0.87934	-20.1427	16.75718	0.00793	0.01073	0.15204	-0.21938
5.00E+08	0.13536	-0.91598	-16.4352	17.43518	0.01023	0.01124	0.10942	-0.2349
6.00E+08	-0.03393	-0.91769	-13.2933	17.45213	0.01229	0.01128	0.07038	-0.24349
7.00E+08	-0.17887	-0.89297	-10.6504	17.02564	0.01393	0.01101	0.03644	-0.24387
8.00E+08	-0.29484	-0.85558	-8.53996	16.36721	0.0153	0.01064	0.01076	-0.2441
9.00E+08	-0.38927	-0.81098	-6.7696	15.60573	0.01634	0.01012	-0.01228	-0.24288
1.00E+09	-0.4656	-0.76476	-5.36755	14.76405	0.01716	0.0097	-0.03097	-0.2393
1.10E+09	-0.52921	-0.71944	-4.2313	13.95409	0.01783	0.0092	-0.04684	-0.23534
1.20E+09	-0.58196	-0.67559	-3.31134	13.19716	0.01846	0.00879	-0.05917	-0.23226
1.30E+09	-0.62496	-0.63192	-2.49966	12.44899	0.01893	0.00825	-0.07328	-0.23094
1.40E+09	-0.66166	-0.59195	-1.84653	11.78412	0.0193	0.00784	-0.08364	-0.23004
1.50E+09	-0.69117	-0.55416	-1.30793	11.15026	0.01955	0.00742	-0.09296	-0.22826
1.60E+09	-0.71596	-0.5188	-0.84939	10.56179	0.01975	0.00701	-0.10169	-0.22696
1.70E+09	-0.73662	-0.48698	-0.47909	10.00766	0.01982	0.00665	-0.10934	-0.22496
1.80E+09	-0.75512	-0.45651	-0.12496	9.51071	0.01991	0.0063	-0.11714	-0.22695
1.90E+09	-0.77157	-0.42898	0.15309	9.02837	0.01987	0.00612	-0.12536	-0.22615
2.00E+09	-0.7865	-0.40241	0.39268	8.60683	0.01993	0.00597	-0.13091	-0.22656
2.10E+09	-0.79926	-0.37678	0.61638	8.20735	0.01998	0.00577	-0.13726	-0.22847
2.20E+09	-0.81032	-0.35292	0.80431	7.839	0.02	0.00558	-0.14297	-0.2299
2.30E+09	-0.82048	-0.32932	0.97501	7.48249	0.02005	0.00535	-0.15032	-0.23182
2.40E+09	-0.82782	-0.3075	1.11863	7.14428	0.0199	0.00511	-0.15747	-0.23248
2.50E+09	-0.83495	-0.2876	1.24052	6.83845	0.01974	0.00504	-0.16276	-0.23348
2.60E+09	-0.84151	-0.26846	1.35502	6.55381	0.01965	0.00497	-0.16797	-0.23611
2.70E+09	-0.84734	-0.25009	1.4657	6.27534	0.01953	0.00496	-0.17506	-0.2396
2.80E+09	-0.85289	-0.23264	1.55177	6.01164	0.0195	0.00493	-0.18194	-0.24149
2.90E+09	-0.85741	-0.21547	1.63052	5.76767	0.01944	0.00486	-0.18872	-0.24356
3.00E+09	-0.86146	-0.19934	1.69921	5.5326	0.01936	0.00483	-0.19566	-0.24536
3.10E+09	-0.86527	-0.1838	1.762	5.30783	0.01923	0.00474	-0.20327	-0.24741
3.20E+09	-0.86831	-0.16914	1.81449	5.09501	0.01901	0.00475	-0.21039	-0.24903
3.30E+09	-0.87139	-0.15503	1.85911	4.89425	0.01885	0.00481	-0.21719	-0.25047
3.40E+09	-0.87419	-0.14165	1.89608	4.70703	0.0188	0.00494	-0.22337	-0.25221
3.50E+09	-0.87733	-0.12853	1.93308	4.52837	0.01881	0.00497	-0.23059	-0.25439
3.60E+09	-0.88011	-0.11517	1.96509	4.35649	0.01873	0.00493	-0.23761	-0.25567
3.70E+09	-0.88223	-0.10206	1.99212	4.19469	0.01864	0.00484	-0.24409	-0.25733
3.80E+09	-0.8838	-0.0893	2.01979	4.04054	0.01844	0.00476	-0.25078	-0.25974

3.90E+09	-0.88517	-0.07762	2.04233	3.88832	0.01818	0.0048	-0.25835	-0.26179
4.00E+09	-0.88673	-0.06618	2.06185	3.74125	0.01799	0.00489	-0.26636	-0.26336
4.10E+09	-0.88792	-0.05495	2.07674	3.60044	0.01778	0.00502	-0.27424	-0.2647
4.20E+09	-0.88926	-0.04401	2.08707	3.46794	0.01766	0.00515	-0.2817	-0.26554
4.30E+09	-0.89032	-0.03307	2.097	3.34133	0.01753	0.00522	-0.28907	-0.26672
4.40E+09	-0.8912	-0.02249	2.10536	3.21786	0.01736	0.00531	-0.29709	-0.26786
4.50E+09	-0.89188	-0.01204	2.11347	3.09843	0.01718	0.0054	-0.30506	-0.26879
4.60E+09	-0.89237	-0.00211	2.11592	2.98252	0.01699	0.00551	-0.31298	-0.26935
4.70E+09	-0.89294	0.00763	2.11821	2.87091	0.01679	0.00567	-0.32106	-0.27002
4.80E+09	-0.89357	0.0173	2.11616	2.76286	0.01663	0.00579	-0.32949	-0.27004
4.90E+09	-0.89401	0.02691	2.1126	2.66352	0.01648	0.00599	-0.33675	-0.26997
5.00E+09	-0.89426	0.03621	2.11238	2.56399	0.01634	0.00612	-0.34497	-0.27107
5.10E+09	-0.89457	0.04545	2.10683	2.46907	0.01616	0.00629	-0.35302	-0.27049
5.20E+09	-0.89469	0.05441	2.10049	2.37669	0.01598	0.00646	-0.36109	-0.26999
5.30E+09	-0.89481	0.06345	2.09605	2.28882	0.01585	0.0066	-0.36928	-0.26986
5.40E+09	-0.89479	0.07232	2.08797	2.20181	0.01565	0.0068	-0.37744	-0.26884
5.50E+09	-0.89457	0.08102	2.07995	2.11784	0.01548	0.00698	-0.38551	-0.26821
5.60E+09	-0.8945	0.08972	2.06842	2.03797	0.01532	0.00722	-0.39341	-0.26659
5.70E+09	-0.89414	0.09823	2.05897	1.9608	0.0152	0.00744	-0.40108	-0.26571
5.80E+09	-0.89392	0.10653	2.04725	1.88578	0.01506	0.00767	-0.4091	-0.26435
5.90E+09	-0.8935	0.11485	2.03554	1.81373	0.01492	0.00788	-0.41699	-0.26298
6.00E+09	-0.89289	0.12296	2.02426	1.74267	0.01475	0.0081	-0.42512	-0.26193
6.10E+09	-0.89242	0.13095	2.01141	1.67349	0.01463	0.00835	-0.43308	-0.2605
6.20E+09	-0.89208	0.139	1.99663	1.60721	0.01452	0.00861	-0.44096	-0.2583
6.30E+09	-0.89157	0.14705	1.98087	1.54186	0.01439	0.0088	-0.4491	-0.25577
6.40E+09	-0.89101	0.15517	1.96611	1.48328	0.01427	0.00906	-0.45589	-0.25357
6.50E+09	-0.89038	0.163	1.94785	1.42152	0.01411	0.00935	-0.4641	-0.25059
6.60E+09	-0.88936	0.17073	1.93442	1.3642	0.014	0.00957	-0.47135	-0.24892
6.70E+09	-0.88858	0.17828	1.91573	1.3115	0.01387	0.00986	-0.4781	-0.24557
6.80E+09	-0.88766	0.18582	1.9013	1.25551	0.01373	0.01011	-0.48603	-0.24355
6.90E+09	-0.88677	0.19335	1.885	1.20472	0.01361	0.01043	-0.49305	-0.24058
7.00E+09	-0.88581	0.20082	1.86885	1.15347	0.0135	0.01076	-0.50037	-0.23749
7.10E+09	-0.88471	0.20841	1.85409	1.10352	0.01345	0.01106	-0.50761	-0.23515
7.20E+09	-0.88358	0.21589	1.83538	1.05525	0.01337	0.01136	-0.51476	-0.23164
7.30E+09	-0.88236	0.22341	1.81765	1.00773	0.01332	0.01166	-0.52179	-0.22862
7.40E+09	-0.88097	0.23078	1.80112	0.96287	0.01322	0.01196	-0.52893	-0.22573
7.50E+09	-0.87976	0.23803	1.7824	0.91977	0.01315	0.01229	-0.53571	-0.22161
7.60E+09	-0.87847	0.24518	1.76452	0.87647	0.01306	0.01262	-0.5432	-0.21815
7.70E+09	-0.87715	0.25235	1.7477	0.83667	0.01303	0.01298	-0.5498	-0.21487
7.80E+09	-0.87576	0.25932	1.72802	0.79656	0.01298	0.01332	-0.55668	-0.2107
7.90E+09	-0.87454	0.26641	1.71013	0.7575	0.01297	0.01367	-0.56351	-0.2068
8.00E+09	-0.87332	0.27366	1.6915	0.72112	0.01295	0.01403	-0.56959	-0.20257
8.10E+09	-0.87144	0.28088	1.67483	0.68378	0.01296	0.01435	-0.57668	-0.19854
8.20E+09	-0.87001	0.28787	1.65642	0.64972	0.01292	0.01477	-0.58261	-0.19397
8.30E+09	-0.86821	0.29492	1.63754	0.6141	0.01294	0.0151	-0.58962	-0.18984
8.40E+09	-0.86634	0.3019	1.61976	0.58045	0.01295	0.01547	-0.59596	-0.18609
8.50E+09	-0.86463	0.30873	1.6003	0.54851	0.01298	0.01589	-0.6021	-0.18143

Frequency	S11		S21		S12		S22	
8.60E+09	-0.86277	0.31572	1.58237	0.5173	0.01308	0.01623	-0.60822	-0.17705
8.70E+09	-0.86106	0.32264	1.56357	0.48729	0.01313	0.0166	-0.6143	-0.17214
8.80E+09	-0.85923	0.32966	1.54583	0.4586	0.0132	0.01698	-0.62009	-0.16748
8.90E+09	-0.8572	0.33685	1.52776	0.42981	0.0133	0.01735	-0.62594	-0.16276
9.00E+09	-0.855	0.34383	1.50948	0.4026	0.01345	0.01768	-0.63128	-0.15806
9.10E+09	-0.85283	0.35073	1.49085	0.37591	0.01355	0.01803	-0.63673	-0.15263
9.20E+09	-0.85055	0.35762	1.47372	0.34869	0.01365	0.01835	-0.64241	-0.14773
9.30E+09	-0.8484	0.36457	1.45562	0.32454	0.01379	0.01873	-0.64725	-0.14207
9.40E+09	-0.846	0.3714	1.43786	0.29894	0.01389	0.01903	-0.65285	-0.13676
9.50E+09	-0.84337	0.37818	1.42042	0.27606	0.01401	0.01937	-0.65748	-0.13191
9.60E+09	-0.84105	0.38452	1.4022	0.2514	0.01404	0.01972	-0.66329	-0.12707
9.70E+09	-0.83841	0.39151	1.38401	0.23088	0.01424	0.02026	-0.66781	-0.12144
9.80E+09	-0.83606	0.39827	1.36741	0.20935	0.01466	0.02064	-0.67281	-0.11579
9.90E+09	-0.83331	0.4053	1.35023	0.1873	0.01491	0.02076	-0.6779	-0.10988
1.00E+10	-0.83056	0.41218	1.33244	0.16691	0.01503	0.02103	-0.68246	-0.104
1.01E+10	-0.82761	0.41894	1.31524	0.14655	0.01521	0.02131	-0.68696	-0.09844
1.02E+10	-0.8246	0.42565	1.29793	0.12789	0.01539	0.02159	-0.69093	-0.09277
1.03E+10	-0.82161	0.43232	1.28092	0.10928	0.01559	0.02189	-0.69502	-0.08696
1.04E+10	-0.81861	0.43902	1.26365	0.09119	0.0158	0.02218	-0.69901	-0.08048
1.05E+10	-0.81563	0.44558	1.24722	0.07361	0.01597	0.02241	-0.70326	-0.07467
1.06E+10	-0.81266	0.45222	1.22997	0.05735	0.01615	0.02274	-0.70713	-0.0683
1.07E+10	-0.80964	0.45882	1.21338	0.04036	0.01637	0.02307	-0.71134	-0.06224
1.08E+10	-0.8065	0.46554	1.19696	0.02545	0.01662	0.02337	-0.71485	-0.05612
1.09E+10	-0.80306	0.47262	1.18086	0.00949	0.01686	0.02364	-0.71856	-0.0496
1.10E+10	-0.79938	0.479	1.16466	-0.00488	0.01718	0.02391	-0.72155	-0.04367
1.11E+10	-0.79596	0.48544	1.14772	-0.01944	0.01737	0.02414	-0.72529	-0.03746
1.12E+10	-0.79254	0.49176	1.13252	-0.03267	0.01761	0.02438	-0.7286	-0.03214
1.13E+10	-0.78922	0.49837	1.11661	-0.04472	0.01788	0.02469	-0.73149	-0.02627
1.14E+10	-0.78476	0.50528	1.10342	-0.05951	0.01821	0.02492	-0.73422	-0.02119
1.15E+10	-0.78118	0.51125	1.08623	-0.07159	0.0184	0.02517	-0.73707	-0.01494
1.16E+10	-0.77731	0.51793	1.07138	-0.08373	0.01867	0.02542	-0.74016	-0.00858
1.17E+10	-0.77344	0.5246	1.05685	-0.09564	0.01894	0.02567	-0.74307	-0.00231
1.18E+10	-0.76944	0.53115	1.04162	-0.10727	0.01916	0.02595	-0.74609	0.00415
1.19E+10	-0.76509	0.53752	1.02727	-0.11804	0.01943	0.02622	-0.74828	0.01009
1.20E+10	-0.76085	0.54362	1.01244	-0.1297	0.01972	0.02659	-0.75094	0.01654
1.21E+10	-0.75682	0.54966	0.99829	-0.13957	0.02017	0.02695	-0.75289	0.02307
1.22E+10	-0.75277	0.55588	0.98307	-0.15	0.02063	0.02718	-0.75536	0.02998
1.23E+10	-0.74861	0.56215	0.96936	-0.15974	0.02106	0.02734	-0.75774	0.03599
1.24E+10	-0.7445	0.56829	0.95458	-0.1685	0.02146	0.02753	-0.76004	0.04229
1.25E+10	-0.74029	0.5744	0.94073	-0.17786	0.0219	0.02766	-0.76241	0.04768
1.26E+10	-0.7359	0.5806	0.92713	-0.18572	0.02229	0.02779	-0.76401	0.05378
1.27E+10	-0.73107	0.58688	0.91363	-0.19493	0.0227	0.02787	-0.76592	0.06021
1.28E+10	-0.72629	0.59315	0.90045	-0.20262	0.02306	0.02798	-0.76762	0.06704
1.29E+10	-0.72142	0.59931	0.88667	-0.21103	0.0234	0.02801	-0.76975	0.07381
1.30E+10	-0.71656	0.60538	0.87358	-0.21824	0.0237	0.02809	-0.77118	0.08001
1.31E+10	-0.71173	0.61115	0.86053	-0.22561	0.02395	0.02824	-0.77269	0.08611
1.32E+10	-0.70678	0.61698	0.84753	-0.23271	0.02428	0.02845	-0.77398	0.09267

1.33E+10	-0.70187	0.62264	0.83476	-0.23968	0.02471	0.02864	-0.77526	0.0994
1.34E+10	-0.69703	0.62845	0.8222	-0.24636	0.02514	0.02877	-0.7766	0.1061
1.35E+10	-0.6922	0.63422	0.80885	-0.25235	0.02553	0.0289	-0.77805	0.11276
1.36E+10	-0.68741	0.63996	0.79639	-0.25791	0.02599	0.02899	-0.77933	0.11823
1.37E+10	-0.68251	0.64549	0.78425	-0.26304	0.02646	0.02902	-0.78051	0.1236
1.38E+10	-0.67739	0.65086	0.77255	-0.26887	0.0269	0.02895	-0.78174	0.1294
1.39E+10	-0.67235	0.65632	0.76065	-0.27404	0.02723	0.02892	-0.78269	0.13581
1.40E+10	-0.66715	0.66188	0.74886	-0.27978	0.02759	0.02887	-0.78399	0.14248
1.41E+10	-0.66172	0.66749	0.73691	-0.28458	0.0278	0.02881	-0.78462	0.14893
1.42E+10	-0.65643	0.6728	0.72501	-0.28914	0.02798	0.02896	-0.78555	0.15483
1.43E+10	-0.65124	0.67813	0.71407	-0.29338	0.02838	0.02912	-0.78595	0.16061
1.44E+10	-0.64597	0.68327	0.70275	-0.29762	0.02877	0.02915	-0.78629	0.16704
1.45E+10	-0.64072	0.68835	0.69204	-0.30214	0.02906	0.02918	-0.78649	0.17364
1.46E+10	-0.6355	0.69372	0.68051	-0.30597	0.02937	0.02934	-0.787	0.18093
1.47E+10	-0.63011	0.69879	0.66951	-0.30975	0.0298	0.02942	-0.78747	0.18773
1.48E+10	-0.62534	0.70382	0.65872	-0.31294	0.03022	0.02946	-0.78785	0.19392
1.49E+10	-0.62035	0.70928	0.64831	-0.31618	0.03066	0.02944	-0.78816	0.20016
1.50E+10	-0.61503	0.71434	0.63772	-0.31956	0.03105	0.02942	-0.78794	0.20632
1.51E+10	-0.60985	0.71963	0.62763	-0.3228	0.03143	0.02931	-0.78793	0.21269
1.52E+10	-0.60406	0.72489	0.61704	-0.3256	0.03159	0.02923	-0.78752	0.21924
1.53E+10	-0.59837	0.7296	0.60659	-0.32818	0.03167	0.02936	-0.78751	0.22507
1.54E+10	-0.59297	0.7344	0.59699	-0.33027	0.0321	0.02977	-0.78748	0.2305
1.55E+10	-0.58782	0.73932	0.58746	-0.33265	0.03272	0.02981	-0.78714	0.2364
1.56E+10	-0.58244	0.74428	0.5782	-0.33503	0.03319	0.02974	-0.78665	0.24244
1.57E+10	-0.5769	0.74923	0.56836	-0.33708	0.0336	0.02972	-0.78622	0.24912
1.58E+10	-0.57122	0.75428	0.55915	-0.33967	0.03402	0.02959	-0.78625	0.25532
1.59E+10	-0.56554	0.75948	0.55003	-0.34135	0.03431	0.02953	-0.78602	0.26177
1.60E+10	-0.55967	0.76446	0.54024	-0.34345	0.03453	0.02946	-0.786	0.26789
1.61E+10	-0.55366	0.76921	0.53144	-0.34506	0.0347	0.02961	-0.78546	0.27404
1.62E+10	-0.54806	0.77362	0.52242	-0.34634	0.03514	0.02989	-0.78504	0.28044
1.63E+10	-0.54266	0.77838	0.51398	-0.34794	0.03574	0.03002	-0.78447	0.2868
1.64E+10	-0.53696	0.78322	0.50503	-0.34941	0.03625	0.03	-0.784	0.29301
1.65E+10	-0.53132	0.78774	0.49665	-0.35063	0.03681	0.0301	-0.78322	0.29813
1.66E+10	-0.5256	0.79251	0.48831	-0.352	0.03762	0.02997	-0.78265	0.304
1.67E+10	-0.51958	0.79698	0.47961	-0.35307	0.03803	0.0296	-0.78136	0.30984
1.68E+10	-0.51344	0.8013	0.47133	-0.35403	0.03839	0.02941	-0.77989	0.3158
1.69E+10	-0.50703	0.8059	0.46317	-0.35516	0.03875	0.02929	-0.77897	0.32238
1.70E+10	-0.50064	0.81016	0.45487	-0.35604	0.0391	0.02911	-0.77791	0.32883
1.71E+10	-0.49443	0.81389	0.44676	-0.35678	0.03945	0.029	-0.77691	0.33487
1.72E+10	-0.48825	0.81781	0.43882	-0.35748	0.03982	0.02888	-0.77549	0.34114
1.73E+10	-0.48215	0.82162	0.43071	-0.35803	0.04011	0.02877	-0.77374	0.34765
1.74E+10	-0.47609	0.82557	0.42272	-0.35834	0.04047	0.02882	-0.77223	0.35425
1.75E+10	-0.46976	0.82938	0.41465	-0.35863	0.04104	0.02873	-0.77073	0.36071
1.76E+10	-0.46364	0.8329	0.40689	-0.35849	0.04149	0.02842	-0.76942	0.3665
1.77E+10	-0.45795	0.83594	0.39925	-0.35812	0.04168	0.02815	-0.76785	0.3718
1.78E+10	-0.45264	0.83905	0.39193	-0.35778	0.04193	0.02809	-0.76604	0.37687
1.79E+10	-0.44695	0.8423	0.38479	-0.35757	0.04226	0.02805	-0.76381	0.38232

1.80E+10	-0.44109	0.84571	0.37765	-0.35727	0.04265	0.02795	-0.76185	0.388
1.81E+10	-0.4351	0.84899	0.37037	-0.3569	0.04305	0.02778	-0.76015	0.39348
1.82E+10	-0.42942	0.8521	0.36372	-0.35606	0.04331	0.02764	-0.75907	0.39839
1.83E+10	-0.42401	0.85498	0.35695	-0.35537	0.04362	0.02757	-0.75803	0.40325
1.84E+10	-0.41844	0.85795	0.3504	-0.35504	0.04401	0.02739	-0.75643	0.40871
1.85E+10	-0.41256	0.86043	0.34335	-0.3544	0.04426	0.02724	-0.75411	0.41448
1.86E+10	-0.40669	0.86318	0.33681	-0.35346	0.04457	0.02715	-0.75204	0.42042
1.87E+10	-0.40107	0.86582	0.33047	-0.35223	0.04484	0.027	-0.75073	0.42579
1.88E+10	-0.39575	0.86861	0.32406	-0.35102	0.04509	0.02693	-0.74985	0.43068
1.89E+10	-0.39099	0.87113	0.31833	-0.34955	0.04534	0.02689	-0.74856	0.43521
1.90E+10	-0.38625	0.87371	0.31288	-0.34844	0.04568	0.0269	-0.74652	0.43926
1.91E+10	-0.38119	0.87569	0.30699	-0.34767	0.04598	0.02681	-0.74409	0.44376
1.92E+10	-0.37669	0.87814	0.30138	-0.34647	0.04632	0.02677	-0.74206	0.44928
1.93E+10	-0.37182	0.88118	0.2957	-0.34545	0.04663	0.02671	-0.74041	0.45518
1.94E+10	-0.36675	0.88416	0.29018	-0.34412	0.04697	0.02675	-0.73898	0.46038
1.95E+10	-0.3621	0.88684	0.28496	-0.34277	0.04735	0.02677	-0.7376	0.46534
1.96E+10	-0.35731	0.88959	0.27976	-0.34172	0.04785	0.02686	-0.73524	0.47057
1.97E+10	-0.35207	0.89164	0.27464	-0.34051	0.04842	0.02679	-0.73201	0.47613
1.98E+10	-0.34683	0.89364	0.26948	-0.33939	0.04905	0.02665	-0.72889	0.48229
1.99E+10	-0.34141	0.89605	0.26417	-0.3384	0.04954	0.02629	-0.72628	0.48843
2.00E+10	-0.3361	0.89834	0.25891	-0.3369	0.04989	0.026	-0.72432	0.49411
2.01E+10	-0.33149	0.90054	0.2542	-0.33537	0.05024	0.02576	-0.72292	0.4992
2.02E+10	-0.32734	0.90282	0.24935	-0.33393	0.0505	0.02555	-0.72094	0.50394
2.03E+10	-0.32281	0.90476	0.24468	-0.33274	0.05078	0.02537	-0.7183	0.50919
2.04E+10	-0.31801	0.90685	0.2397	-0.33144	0.05104	0.02526	-0.71483	0.51496
2.05E+10	-0.31292	0.9091	0.23466	-0.32987	0.05133	0.02519	-0.7114	0.52068
2.06E+10	-0.30787	0.91102	0.22987	-0.32818	0.05174	0.0251	-0.70861	0.52576
2.07E+10	-0.30317	0.91284	0.22531	-0.32629	0.05218	0.02498	-0.70604	0.53056
2.08E+10	-0.29932	0.91423	0.22112	-0.32444	0.05261	0.02485	-0.70317	0.53459
2.09E+10	-0.29539	0.91528	0.21693	-0.32279	0.05303	0.02463	-0.70009	0.53927
2.10E+10	-0.29142	0.91676	0.21269	-0.32118	0.0534	0.02443	-0.69688	0.54425
2.11E+10	-0.28683	0.91815	0.20819	-0.31943	0.05375	0.02417	-0.69374	0.54949
2.12E+10	-0.2823	0.91984	0.20404	-0.31746	0.05405	0.02398	-0.69136	0.55408
2.13E+10	-0.27835	0.92147	0.20007	-0.3153	0.05439	0.02383	-0.6896	0.55859
2.14E+10	-0.27456	0.92293	0.19632	-0.31339	0.05476	0.02363	-0.68779	0.56275
2.15E+10	-0.27069	0.92437	0.19249	-0.31155	0.05513	0.02341	-0.68509	0.56681
2.16E+10	-0.26661	0.92534	0.18865	-0.30985	0.05545	0.02314	-0.68221	0.5711
2.17E+10	-0.26259	0.92647	0.18491	-0.30785	0.05574	0.02293	-0.67909	0.5756
2.18E+10	-0.25846	0.92763	0.18106	-0.30584	0.05599	0.02269	-0.67638	0.58012
2.19E+10	-0.25464	0.92867	0.17762	-0.30376	0.05628	0.02255	-0.67398	0.58451
2.20E+10	-0.25155	0.92957	0.17429	-0.30163	0.05664	0.02242	-0.67124	0.5889
2.21E+10	-0.24872	0.93025	0.17102	-0.2997	0.05698	0.0222	-0.66839	0.59322
2.22E+10	-0.24566	0.93108	0.1678	-0.29795	0.05727	0.02194	-0.66455	0.59784
2.23E+10	-0.24198	0.93216	0.16435	-0.29594	0.05754	0.02178	-0.66044	0.60251
2.24E+10	-0.23797	0.93251	0.161	-0.29378	0.05785	0.02164	-0.6567	0.60705
2.25E+10	-0.23401	0.93198	0.15806	-0.29145	0.05824	0.02146	-0.65346	0.61142
2.26E+10	-0.23237	0.93242	0.15535	-0.28953	0.05866	0.02121	-0.65136	0.61544

2.27E+10	-0.22983	0.93355	0.15255	-0.28752	0.059	0.02092	-0.64885	0.61957
2.28E+10	-0.22714	0.93436	0.14969	-0.2856	0.05934	0.0206	-0.64591	0.62343
2.29E+10	-0.22433	0.93491	0.14708	-0.28401	0.05967	0.02025	-0.64257	0.62741
2.30E+10	-0.2216	0.93573	0.14439	-0.28201	0.05997	0.01988	-0.6387	0.63193
2.31E+10	-0.21878	0.93641	0.14155	-0.28039	0.06017	0.01943	-0.63497	0.6363
2.32E+10	-0.21587	0.93743	0.13875	-0.2787	0.06023	0.01895	-0.63148	0.64066
2.33E+10	-0.21335	0.93811	0.13608	-0.27674	0.06006	0.01864	-0.62829	0.64434
2.34E+10	-0.2109	0.9387	0.13317	-0.27488	0.05995	0.0186	-0.62539	0.64796
2.35E+10	-0.2089	0.93905	0.13061	-0.27283	0.06	0.0187	-0.62196	0.65178
2.36E+10	-0.20652	0.93921	0.12789	-0.27097	0.06019	0.01873	-0.61773	0.65652
2.37E+10	-0.20388	0.93913	0.12512	-0.2688	0.0604	0.01879	-0.61304	0.66118
2.38E+10	-0.2016	0.93893	0.12261	-0.26652	0.06074	0.01882	-0.60903	0.6659
2.39E+10	-0.1996	0.93948	0.12031	-0.26426	0.06114	0.01874	-0.6058	0.66995
2.40E+10	-0.198	0.94004	0.1183	-0.26197	0.06151	0.01864	-0.60272	0.67305
2.41E+10	-0.19689	0.94076	0.11653	-0.26001	0.06195	0.01849	-0.59984	0.67573
2.42E+10	-0.19541	0.94196	0.11484	-0.25849	0.06237	0.01826	-0.59629	0.67874
2.43E+10	-0.19298	0.94279	0.11272	-0.25691	0.06263	0.01797	-0.5926	0.68239
2.44E+10	-0.18988	0.94348	0.11037	-0.25514	0.06292	0.0177	-0.5888	0.68628
2.45E+10	-0.18678	0.94436	0.10813	-0.25331	0.06319	0.01742	-0.58599	0.69024
2.46E+10	-0.18412	0.94471	0.10613	-0.25149	0.06335	0.01712	-0.58392	0.69393
2.47E+10	-0.18211	0.94439	0.10405	-0.24969	0.06347	0.0169	-0.58077	0.69788
2.48E+10	-0.18049	0.94402	0.10183	-0.24805	0.06367	0.0167	-0.57597	0.7024
2.49E+10	-0.17895	0.94318	0.09963	-0.24621	0.06381	0.01659	-0.56993	0.70681
2.50E+10	-0.17716	0.94292	0.09761	-0.24416	0.06403	0.01648	-0.5644	0.71108
2.51E+10	-0.17504	0.9431	0.09551	-0.24205	0.06426	0.01637	-0.56029	0.71489
2.52E+10	-0.1733	0.94355	0.09391	-0.23989	0.06456	0.01621	-0.55779	0.71764
2.53E+10	-0.17278	0.94378	0.09246	-0.23788	0.06473	0.01612	-0.55579	0.71972
2.54E+10	-0.17208	0.94433	0.09105	-0.23604	0.06507	0.01608	-0.55296	0.72234
2.55E+10	-0.17043	0.94488	0.08953	-0.23458	0.06554	0.01587	-0.54827	0.72643
2.56E+10	-0.16742	0.94478	0.08777	-0.23314	0.06599	0.01547	-0.54232	0.73138
2.57E+10	-0.16401	0.94469	0.08587	-0.23134	0.06628	0.015	-0.53713	0.73549
2.58E+10	-0.16183	0.94466	0.08435	-0.22952	0.06642	0.01458	-0.53411	0.73838
2.59E+10	-0.16045	0.94494	0.08303	-0.22799	0.06652	0.01414	-0.53147	0.74055
2.60E+10	-0.15955	0.94509	0.0815	-0.22666	0.06646	0.01382	-0.52802	0.74291

Appendix D – Rogers RO3000 Series Materials Data Sheet



RO3000® Series Circuit Materials RO3003™, RO3006™, RO3010™ and RO3035™ High Frequency Laminates

RO3000® high frequency circuit materials are ceramic-filled PTFE composites intended for use in commercial microwave and RF applications. This family of products was designed to offer exceptional electrical and mechanical stability at competitive prices.

RO3000 series laminates are ceramic-filled PTFE based circuit materials with mechanical properties that are consistent regardless of the dielectric constant selected. This allows the designer to develop multi-layer board designs that use different dielectric constant materials for individual layers, without encountering warpage or reliability problems.

RO3000 materials exhibit a coefficient of thermal expansion (CTE) in the X and Y axis of 17 ppm/°C. This expansion coefficient is matched to that of copper, which allows the material to exhibit excellent dimensional stability, with typical etch shrinkage (after etch and bake) of less than 0.5 mils per inch. The Z-axis CTE is 24 ppm/°C, which provides exceptional plated through-hole reliability, even in severe thermal environments. The dielectric constant versus temperature for RO3003™ and RO3035™ materials is very stable (Chart 1).

RO3000 series laminates can be fabricated into printed circuit boards using standard PTFE circuit board processing techniques, with minor modifications as described in the application note "Fabrication Guidelines for RO3000 Series High Frequency Circuit Materials."

RO3000 laminates are manufactured under an ISO 9001 certified system.



Advanced Connectivity Solutions
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Data Sheet

FEATURES AND BENEFITS:

Low dielectric loss (RO3003™ laminates)

- Laminates can be used in applications up to 77 GHz.

Excellent mechanical properties versus temperature

- Reliable stripline and multi-layer board constructions.

Uniform mechanical properties for a range of dielectric constants

- Ideal for multi-layer board designs with a range of dielectric constants
- Suitable for use with epoxy glass multi-layer board hybrid designs

Stable dielectric constant versus temperature and frequency (RO3003 laminates)

- Ideal for band pass filters, microstrip patch antennas, and voltage controlled oscillators.

Low in-plane expansion coefficient (match to copper)

- Allows for more reliable surface mounted assemblies
- Ideal for applications sensitive to temperature change
- Excellent dimensional stability

Volume manufacturing process

- Economical laminate pricing

SOME TYPICAL APPLICATIONS:

- Automotive radar applications
- Global positioning satellite antennas
- Cellular telecommunications systems - power amplifiers and antennas
- Patch antenna for wireless communications
- Direct broadcast satellites
- Datalink on cable systems
- Remote meter readers
- Power backplanes

Chart 1: RO3003 and RO3035 Laminate Dielectric Constant vs. Temperature

The data in Chart 1 demonstrates the excellent stability of dielectric constant over temperature for RO3003 & RO3035 laminates, including the elimination of the step change in dielectric constant, which occurs near room temperature with PTFE glass materials.

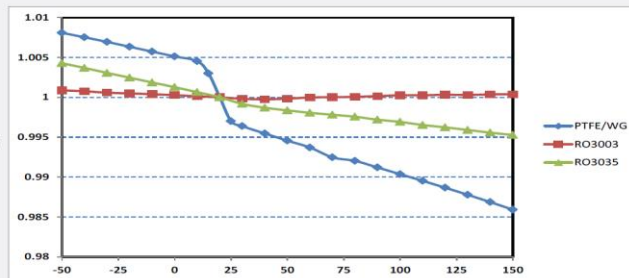


Chart 2: RO3003 and RO3035 Dissipation Factor

The data in Chart 2 shows the distribution of dissipation factor for RO3003 and RO3035 materials.

Test Method: IPC-TM-650 2.5.5.5
Condition: 10 GHz 23° C

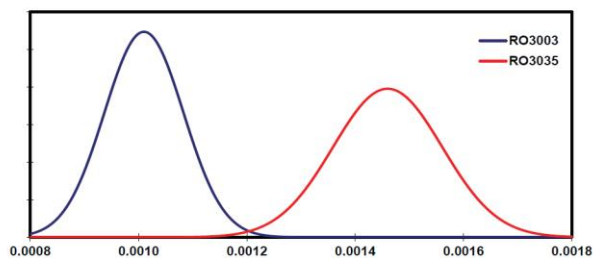
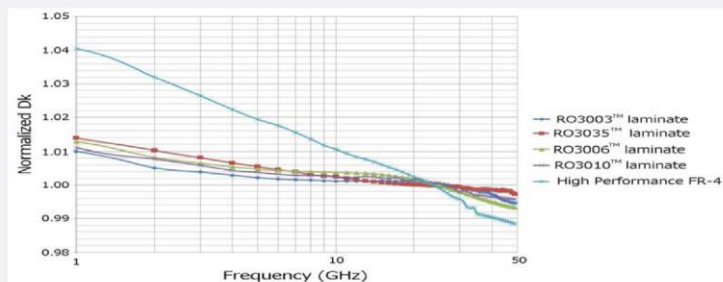

Chart 3: Normalized Dk vs. Frequency using microstrip differential phase length method
50 ohm microstrip circuits based on ~20mil thick laminates

Chart 3 demonstrates the stability of dielectric constant for RO3000 series products over frequency. This stability simplifies the design of broadband components as well as allowing the materials to be used in a wide range of applications over a very broad range of frequencies.



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Property	Typical Value ⁽¹⁾				Direction	Unit	Condition	Test Method
	RO3003	RO3035	RO3006	RO3010				
Dielectric Constant, ϵ_r , Process	3.00 ± 0.04	3.50 ± 0.05	6.15 ± 0.15	10.2 ± 0.30	Z	-	10 GHz 23°C	IPC-TM-650 2.5.5.5 Clamped Stripline
⁽²⁾ Dielectric Constant, ϵ_r , Design	3.00	3.60	6.50	11.20	Z	-	8 GHz - 40 GHz	Differential Phase Length Method
Dissipation Factor, tan δ	0.0010	0.0015	0.0020	0.0022	Z	-	10 GHz 23°C	IPC-TM-650 2.5.5.5
Thermal Coefficient of ϵ_r	-3	-45	-262	-395	Z	ppm/°C	10 GHz -50 to 150°C	IPC-TM-650 2.5.5.5
Dimensional Stability	-0.06 0.07	-0.11 0.11	-0.27 -0.15	-0.35 -0.31	X Y	mm/m	COND A	IPC TM-650 2.2.4
Volume Resistivity	10 ⁷	10 ⁷	10 ⁵	10 ⁵		MΩ•cm	COND A	IPC 2.5.17.1
Surface Resistivity	10 ⁷	10 ⁷	10 ⁵	10 ⁵		MΩ	COND A	IPC 2.5.17.1
Tensile Modulus	930 823	1025 1006	1498 1293	1902 1934	X Y	MPa	23°C	ASTM D638
Moisture Absorption	0.04	0.04	0.02	0.05	-	%	D48/50	IPC-TM-650 2.6.2.1
Specific Heat	0.9		0.86	0.8		J/g/K		Calculated
Thermal Conductivity	0.50	0.50	0.79	0.95	-	W/m/K	50°C	ASTM D5470
Coefficient of Thermal Expansion	17 16 25	17 17 24	17 17 24	13 11 16	X Y Z	ppm/°C	-55 to 288°C	ASTM D3386-94
Td	500	500	500	500		°C TGA		ASTM D3850
Density	2.1	2.1	2.6	2.8		gm/cm ³	23°C	ASTM D792
Copper Peel Strength	12.7	10.2	7.1	9.4		lb/in	1 oz. EDC After Solder Float	IPC-TM-2.4.8
Flammability	V-0	V-0	V-0	V-0				UL 94
Lead Free Process Compatible	YES	YES	YES	YES				

NOTES:

(1) Typical values are a representation of an average value for the population of the property. For specification values contact Rogers Corporation.

(2) The design Dk is an average number from several different tested lots of material and on the most common thickness/s. If more detailed information is required, please contact Rogers Corporation or refer to Rogers' technical papers in the Roger Technology Support Hub available at <http://www.rogerscorp.com/acm/technology>.

Standard Thickness	Standard Panel Size	Available Copper Cladding
RO3003/RO3035: 0.005" (0.13mm) 0.010" (0.25mm) 0.020" (0.50mm) 0.030" (0.75mm) 0.060" (1.52mm) RO3006/RO3010: 0.005" (0.13mm) 0.010" (0.25mm) 0.025" (0.64mm) 0.050" (1.28mm)	12" X 18" (305 X 457mm) 24" X 18" (610 X 457mm)	¼ oz. (9µm) electrodeposited copper foil (HQ/HQ)
		½ oz. (17µm) electrodeposited copper foil (HH/HH)
		1 oz. (35µm) electrodeposited copper foil (H1/H1)
		2 oz. (70µm) electrodeposited copper foil (H2/H2)
		½ oz. (17µm) reverse treated electrodeposited cu foil (SH/SH)
		1 oz. (35µm) reverse treated electrodeposited cu foil (S1/S1)
		2 oz. (70µm) reverse treated electrodeposited cu foil (S2/S2)
		RO3003 & RO3035 laminates also available with ½, 1 and 2 oz. rolled copper foil
		RO3003 5 mil thick laminates also available with 6 oz. rolled copper foil and 0.040" rolled copper plate
		Other claddings may be available. Contact customer service.

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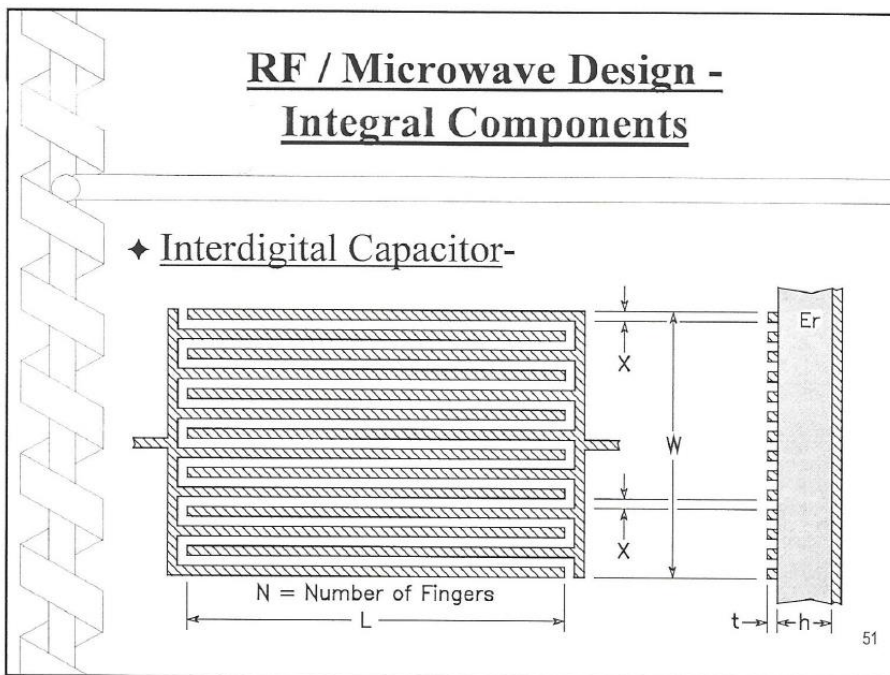
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Appendix E – Interdigital Capacitor Structure and Design Equation



RF / Microwave Design -
Integral Components

♦ Interdigital Capacitor

$$C_2 = \frac{\epsilon_r + 1.0}{w} L \left[(N - 3.0) A1 + A2 \right] \text{ (pF / in)}$$

$$A1 = \left[0.3349057 - 0.15287116 \left(\frac{t}{X} \right) \right]^2$$

$$A2 = \left[0.50133101 - 0.22820444 \left(\frac{t}{X} \right) \right]^2$$

(Equation valid for $h > w/N$)

52

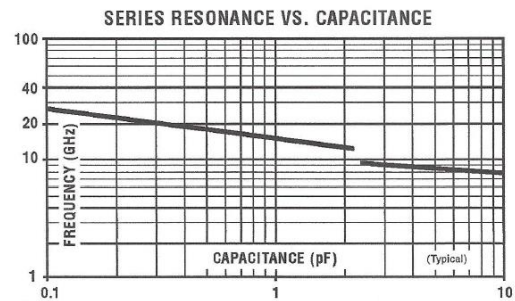
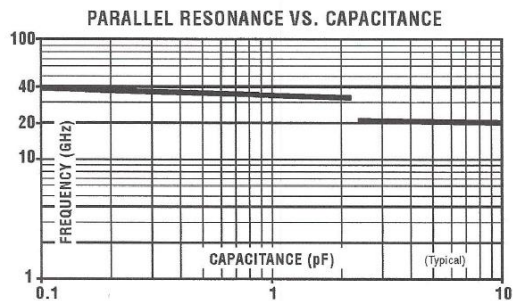
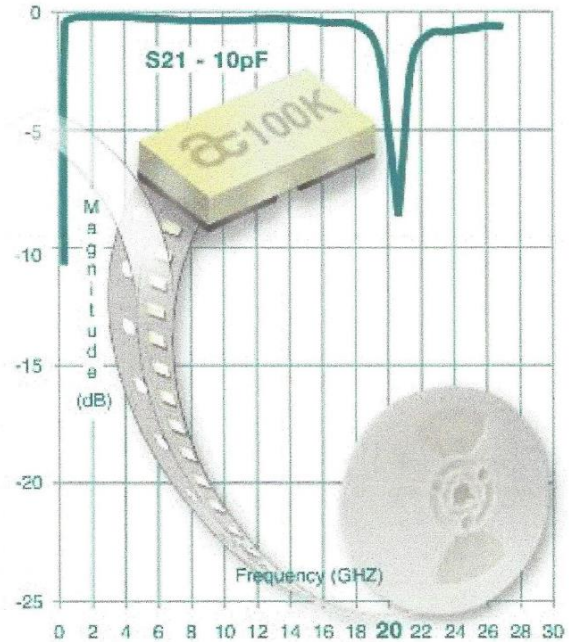
Appendix F – ATC 500 S Series SMT Capacitors Datasheet

ATC 500 S Series BMC Broadband Microwave Millimeter-wave 0603 NPO SMT Capacitors

- Low Insertion Loss
- Ultra High Self Resonance
- Surface Mountable
- Rugged Construction

ATC introduces the new 500 S Series Broadband Microwave Capacitor (BMC). This unique, patented component greatly exceeds both multilayer and single layer capacitor performance. It delivers extremely low insertion loss with ultra-high self resonance performance, in a rugged, laser-marked package compatible with automatic SMT manufacturing.

Functional applications include Broadband (Bypass, Coupling, Feedback, Impedance Matching, D.C. Blocking) and Tuning.



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ATC 001-883
Rev. F; 2/08

ATC 500 S Series Broadband 0603 NPO Capacitors

ELECTRICAL AND MECHANICAL SPECIFICATIONS

QUALITY FACTOR (Q):

Greater than 1,000 @ 1 MHz.

TEMPERATURE COEFFICIENT OF CAPACITANCE (T.C.):

0 $\pm$ 30 PPM/ $^{\circ}$ C (-55° C to $+125^{\circ}$ C) (0.1 pF to 2.2 pF)

0 $\pm$ 60 PPM/ $^{\circ}$ C (-55° C to $+125^{\circ}$ C) (2.4 pF to 10 pF)

INSULATION RESISTANCE (IR):

0.1 pF to 10 pF:

10⁵ Megohms min. @ $+25^{\circ}$ C @ rated WVDC.

10⁴ Megohms min. @ $+125^{\circ}$ C @ rated WVDC.

WORKING VOLTAGE (WVDC):

100 WVDC (0.1 pF to 4.7 pF)

50 WVDC (5.1 pF to 10 pF)

DIELECTRIC WITHSTANDING VOLTAGE (DWV):

200% of rated WVDC for 5 secs.

OPERATING TEMPERATURE RANGE:

From -55° C to $+125^{\circ}$ C (No derating of working voltage).

TERMINATION: Chip style suitable for surface mounting.

Platinum with gold flash.

Die Shear Strength Test per MIL-STD-883, Method 2019.

ENVIRONMENTAL TESTS

ATC 500 S Series Broadband Microwave Capacitors are designed and manufactured to meet and exceed the applicable requirements of MIL-C-55681.

THERMAL SHOCK:

MIL-STD-202, Method 107, Condition A.

MOISTURE RESISTANCE:

MIL-STD-202, Method 106.

LOW VOLTAGE HUMIDITY:

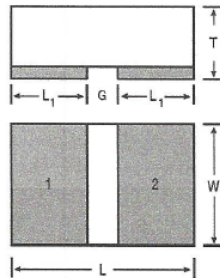
MIL-STD-202, Method 103, Condition A, with 1.5 Volts D.C. applied while subjected to an environment of 85° C with 85% relative humidity for 250 $\pm$ 12 hours.

LIFE TEST:

MIL-STD-202, Method 108, for 2000 hours, at 125° C.

200% WVDC applied.

ATC 500 S SERIES BROADBAND 0603 CAPACITORS: MECHANICAL CONFIGURATIONS



MECHANICAL DIMENSIONS – INCHES (mm)				
Length (L)	L ₁	Width (W)	Thickness (T)	Gap (G)
.060 $\pm$.005 (1.52 $\pm$.127)	.025 (.635) nom.	.030 $\pm$.005 (.762 $\pm$.127)	.024 (.609) max.	.010 (.254) min.

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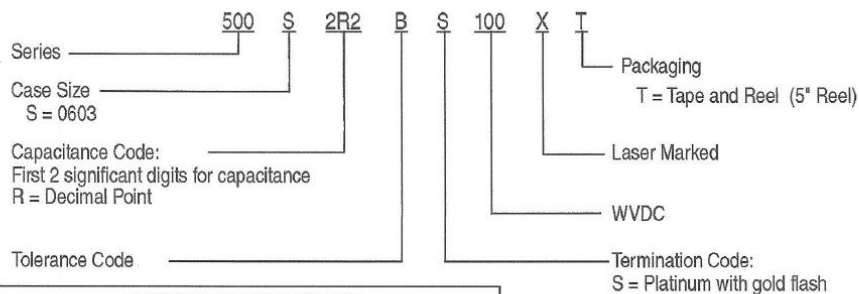
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ATC 500 S Series Broadband 0603 NPO Capacitors

CAPACITANCE VALUES

CAP. CODE	CAP. (pF)	TOL.	RATED WVDC	CAP. CODE	CAP. (pF)	TOL.	RATED WVDC	CAP. CODE	CAP. (pF)	TOL.	RATED WVDC	CAP. CODE	CAP. (pF)	TOL.	RATED WVDC
0R1	0.1	A, B, C, D	100	1R1	1.1	B, C, D	100	3R0	3.0	B, C, D	100	6R8	6.8	C, J, K, M	50
0R2	0.2			1R2	1.2			3R3	3.3			7R5	7.5		
0R3	0.3			1R3	1.3			3R6	3.6			8R2	8.2		
0R4	0.4			1R5	1.5			3R9	3.9			9R1	9.1		
0R5	0.5			1R6	1.6			4R3	4.3			100	10		
0R6	0.6			1R8	1.8			4R7	4.7		50		G, J, K, M		
0R7	0.7			2R0	2.0			5R1	5.1						
0R8	0.8			2R2	2.2			5R6	5.6						
0R9	0.9			2R4	2.4			6R2	6.2						
1R0	1.0			2R7	2.7										

ATC 500 SERIES BMC PART NUMBER CODE



CAPACITANCE TOLERANCE							
Code	A	B	C	D	G	J	M
Tol.	±0.05 pF	±0.1 pF	±0.25 pF	±0.5 pF	±2%	±5%	±20%

The above part number refers to a 500S Series (case size S) 2.2 pF capacitor, B tolerance (±0.1 pF), Termination Code S (Platinum with gold flash), 100 WVDC, with marking and 5" tape and reel packaging.

ATC accepts orders for our parts using designations *with* or *without* the "ATC" prefix. Both methods of defining the part number are equivalent, i.e., part numbers referenced with the "ATC" prefix are interchangeable to parts referenced without the "ATC" prefix. Customers are free to use either in specifying or procuring parts from American Technical Ceramics.

For additional information and catalogs contact your ATC representative or call direct at (+1-631) 622-4700. Consult factory for additional performance data.

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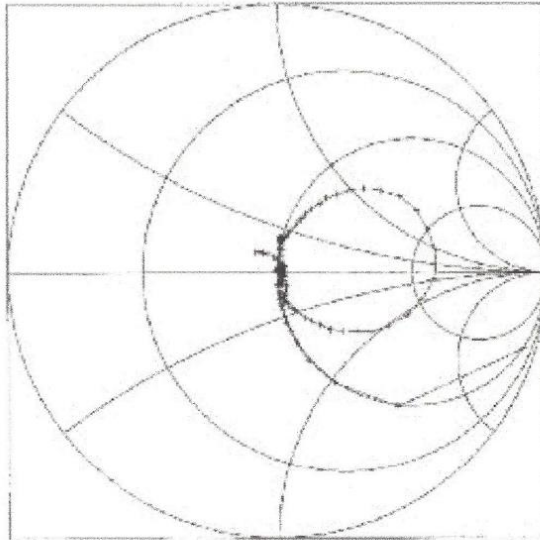
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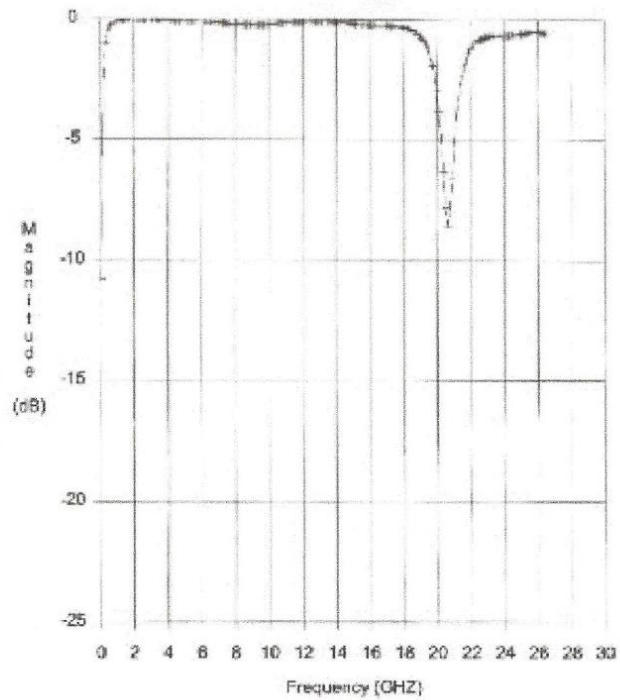
ATC 500 S Series Broadband 0603 NPO Capacitors

S11 - 10 pf



Frequency Range: 50 MHZ to 26.5 GHZ

S21 - 10 pF



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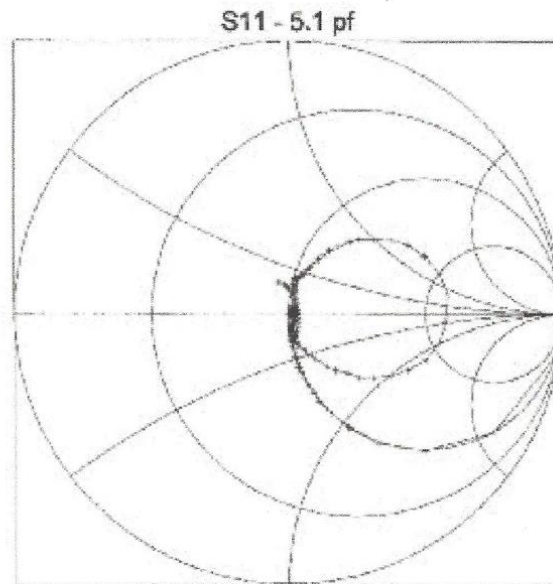
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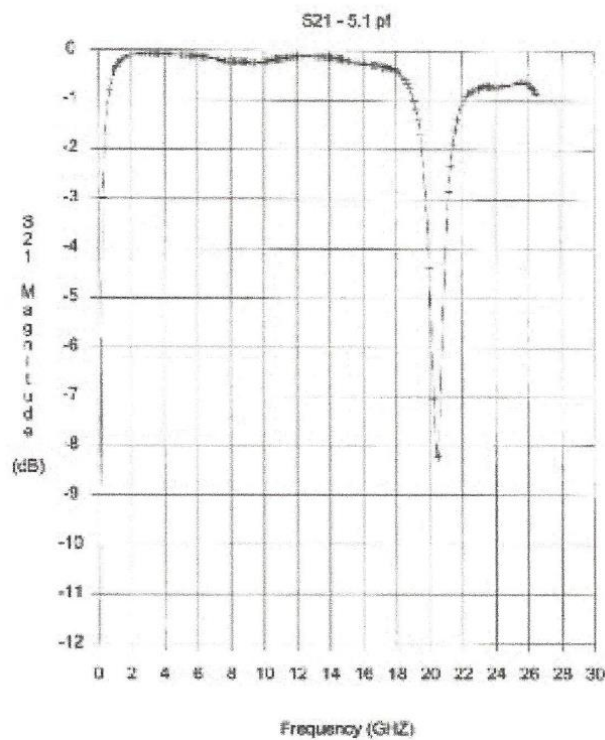
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ATC 500 S Series Broadband 0603 NPO Capacitors



Frequency Range: 50 MHz to 26.5 GHz



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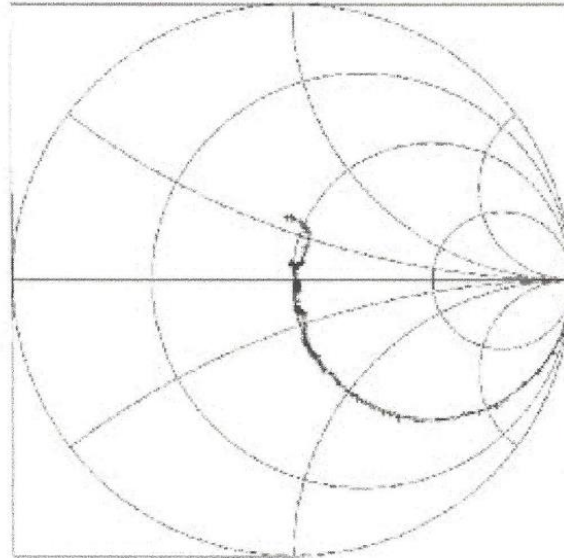
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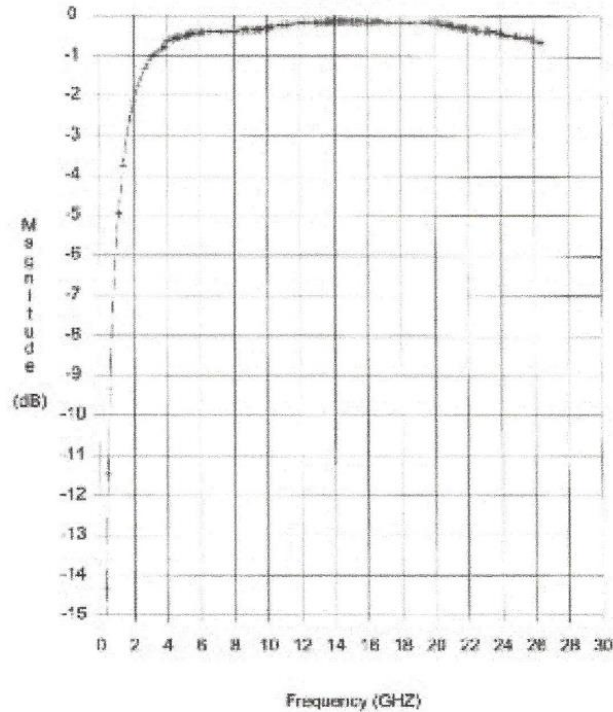
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S11 - 1 pf



Frequency Range: 50 MHZ to 26.5 GHZ

S21 - 1 pf



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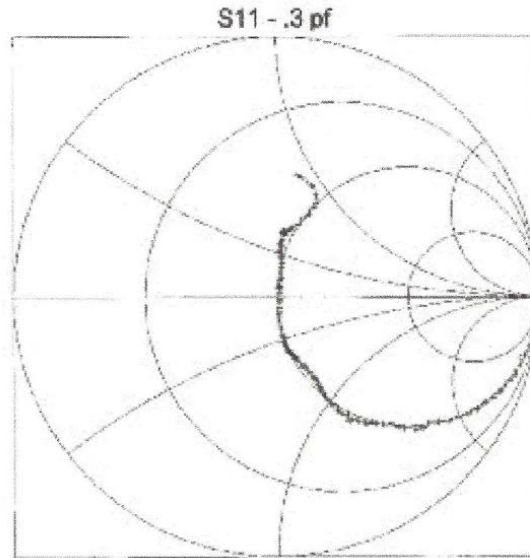
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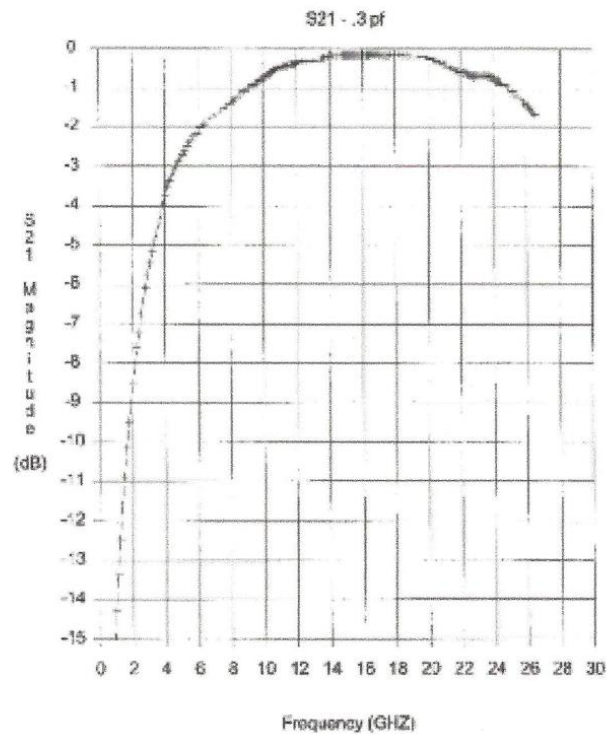
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ATC 500 S Series Broadband 0603 NPO Capacitors



Frequency Range: 50 MHZ to 26.5 GHZ



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